Projector-Resolved Seed Calculus VI

Transfer Semigroups: Stochastic, Bayesian, Fractional, and PDE Realizations

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Fifth Lean-checked formalization block; all 33 results covered

Abstract

This paper develops the transfer-semigroup realization of the projector-resolved seed calculus. A seed is propagated by a one-parameter semigroup, a Markov kernel, a normalized Bayesian transfer, a fractional spectral power, a Green operator, or a potential-weighted Feynman–Kac operator. Exact packet projectors resolve every transfer into input-output blocks, and the semigroup law becomes block convolution over intermediate packets. Coarse and fine generators are compared by a refinement defect. A Duhamel identity proves that this generator-level defect is exactly the source of the corresponding semigroup defect and gives a quantitative stability bound. In the stochastic realization, point-mass seeds generate random walks and continuous-time Markov evolution; exact lumpability is the vanishing of the same refinement defect. Conditional expectation is treated as a weighted orthogonal packet projector, with nested partitions producing martingale-difference packets. Bayesian conditioning is a normalized nonlinear operator. Its independent-data composition law, packet-evidence decomposition, stability bound, conjugate updates, and continuous likelihood flow are proved explicitly. The Bayesian refinement conjecture is resolved in finite dimensions: for strictly positive likelihoods and at least two coarse packets, exact commutation for every prior is equivalent to blockwise-constant likelihoods up to global scale, and every failure factors through an explicit information-loss packet. Fractional powers are constructed packetwise through the spectral projectors of a positive operator, and the square-root semigroup is recovered by subordination. The fractional-naturality conjecture is also resolved for bounded nonnegative self-adjoint refinement systems: the fractional intertwining defect has an exact resolvent integral, a gap-free Hölder certificate, a sharper common-gap estimate, and a Duhamel semigroup bound. Green kernels, finite-difference Poisson problems, and second-order low-mode convergence provide the PDE realization. A finite-state Feynman–Kac formula is proved by path enumeration. Exact Schur–Feshbach elimination shows that discarded latent packets return as a frequency-dependent self-energy and, in time coordinates, as a memory kernel. Survival and hazard functions are generated by transient semigroups. Nineteen code-generated figures, machine-readable results, the original thirty-seven automated tests, and seventeen added boundary-case checks accompany this reviewed edition. Five checked Lean blocks cover all 33 theorem-like results in Appendix A. Finite-dimensional theorems, classical infinite-dimensional extensions, numerical certificates, and open continuum claims are kept separate.

Scope and claim boundary. The paper proves the stated finite-dimensional semigroup, packet, Markov, Bayesian, conditional-expectation, fractional spectral, Green-kernel, Feynman–Kac, Schur–Feshbach, and survival identities. It also proves quantitative refinement-defect formulas under explicit norm hypotheses, including exact transfer of a bounded generator defect to every fractional power $0 < \alpha < 1$ for bounded nonnegative self-adjoint generators. It does not claim that an arbitrary infinite-dimensional operator generates a semigroup, that every stochastic process admits a finite packet representation, that every posterior family is conjugate, that finite Bayesian sufficiency automatically extends to infinite disintegrations, or that every PDE continuum limit follows from a finite test. Domain, closability, sectoriality, measurability, tightness, and compactness hypotheses remain necessary in their respective infinite-dimensional settings. Classical sources are retained in the separate end-of-series reference ledger rather than reproduced in this paper.

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1 The transfer-semigroup realization

Let X be a finite-dimensional real or complex vector space, let $A \in \mathrm{End}(X)$, and let $s \in X$ be a primitive seed. The basic continuous transfer is

$$T(t) = e^{tA}, \qquad t \geq 0,$$

while a positive operator L is more naturally written

$$H(t) = e^{-tL}.$$

Both satisfy the semigroup law. The sign convention changes with the domain: Markov generators and evolution equations are usually written with e^{tA}; positive diffusion and elliptic operators are usually written with e^{-tL}.

The common operator chain is

$$\boxed{\text{seed} \longrightarrow \text{transfer} \longrightarrow \text{project} \longrightarrow \text{condition} \longrightarrow \text{refine} \longrightarrow \text{complete} \longrightarrow \text{reconstruct}.}$$

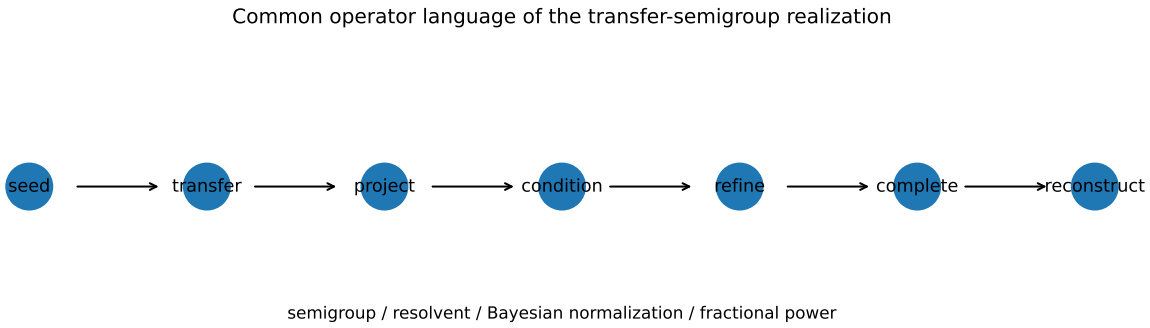


Figure 1: The common operator language used throughout the paper. Semigroups, resolvents, normalized Bayesian maps, and fractional powers are distinct realizations of the same transfer stage.

Definition 1.1 (Transfer-semigroup seed system)**.** A finite transfer-semigroup seed system is a tuple

$$\mathfrak{S} = (X, s, A, \{P_\alpha\}_{\alpha \in I}, \tau),$$

where X is the state space, s is the seed, A is a transfer generator, the P_α are pairwise orthogonal idempotents summing to the identity, and τ specifies the topology in which a completion or limit is interpreted.

Theorem 1.2 (Finite semigroup law)**.** *For every matrix A and all $s, t \in \mathbb{R}$,*

$$\boxed{e^{(s+t)A} = e^{sA} e^{tA}.}$$

Moreover, $T(0) = I$ and $T'(t) = AT(t) = T(t)A$.

Proof. Both exponentials are absolutely convergent matrix power series. Since every power of A commutes with every other power, the Cauchy product gives

$$e^{sA} e^{tA} = \sum_{m,n \geq 0} \frac{s^m t^n}{m! n!} A^{m+n} = \sum_{r \geq 0} \frac{(s+t)^r}{r!} A^r.$$

Termwise differentiation is valid in finite dimension and gives the derivative identities. □

Boundary 1.3 (Infinite-dimensional generators). For an unbounded operator, writing e^{tA} is not a definition unless a semigroup-generation theorem applies. The finite matrix identities in this paper remain algebraically correct models, but a continuum realization must state the operator domain, closure, growth bound, and topology of convergence.

2 Packet blocks of a semigroup

Let

$$P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \qquad \sum_{\alpha \in I} P_\alpha = I.$$

For any operator B, define

$$B_{\beta\alpha} = P_\beta B P_\alpha.$$

Then

$$B = \sum_{\alpha,\beta} B_{\beta\alpha}.$$

For the semigroup, write

$$T_{\beta\alpha}(t) = P_\beta e^{tA} P_\alpha.$$

Theorem 2.1 (Packet semigroup convolution). *For every s, t and every pair of packets α, β,*

$$\boxed{T_{\beta\alpha}(s+t) = \sum_{\gamma \in I} T_{\beta\gamma}(s) T_{\gamma\alpha}(t).}$$

Thus the semigroup law becomes ordinary matrix multiplication in the packet indices.

Proof. Insert the resolution of the identity between the two semigroup factors:

$$\begin{aligned} P_\beta e^{(s+t)A} P_\alpha &= P_\beta e^{sA} \left(\sum_\gamma P_\gamma \right) e^{tA} P_\alpha \\ &= \sum_\gamma (P_\beta e^{sA} P_\gamma)(P_\gamma e^{tA} P_\alpha). \end{aligned}$$

□

Corollary 2.2 (Reducing packets). *If $P_\beta A P_\alpha = 0$ for every $\beta \neq \alpha$, then*

$$P_\beta e^{tA} P_\alpha = 0 \quad (\beta \neq \alpha)$$

for every t. Each packet evolves independently under $P_\alpha A P_\alpha$.

Proof. The hypothesis makes A block diagonal. Every power A^m is block diagonal with respect to the same projectors, and therefore so is the exponential series. □

Remark 2.3 (Path expansion). Expanding the exponential and inserting $I = \sum_\alpha P_\alpha$ between every pair of generators gives

$$P_{\alpha_m} A^m P_{\alpha_0} = \sum_{\alpha_1, \ldots, \alpha_{m-1}} P_{\alpha_m} A P_{\alpha_{m-1}} \cdots P_{\alpha_1} A P_{\alpha_0}.$$

Every coefficient is therefore a sum over packet-transfer paths. The numerical library verifies that truncating this path series converges to the ordinary matrix exponential.

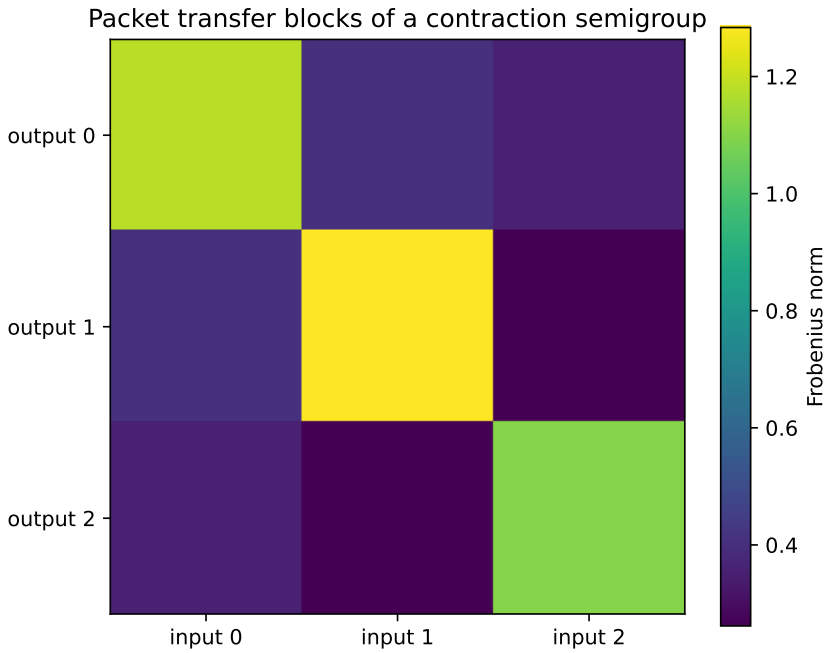


Figure 2: Norms of the packet blocks $P_\beta e^{tA} P_\alpha$. Off-diagonal entries quantify transfer between packets.

3 Refinement defects and the Duhamel identity

Let X_f and X_c be fine and coarse spaces, with generators A_f and A_c, and let

$$C : X_f \to X_c$$

be a coarsening map. Define the generator-level refinement defect

$$\boxed{\Delta = CA_f - A_c C.}$$

The corresponding semigroup defect is

$$\mathfrak{D}(t) = Ce^{tA_f} - e^{tA_c} C.$$

Theorem 3.1 (Exact Duhamel refinement identity)**.** *For every $t \geq 0$,*

$$\boxed{Ce^{tA_f} - e^{tA_c} C = \int_0^t e^{(t-r)A_c} \Delta e^{rA_f}\, dr.}$$

In particular, $\Delta = 0$ implies exact semigroup intertwining at every time.

Proof. Set

$$F(r) = e^{(t-r)A_c} C e^{rA_f}.$$

Then

$$F'(r) = e^{(t-r)A_c} (CA_f - A_c C) e^{rA_f}.$$

Integrating from 0 to t gives $F(t) - F(0) = Ce^{tA_f} - e^{tA_c} C$ and proves the formula. □

Corollary 3.2 (Norm bound)**.** *Suppose*

$$\left\|e^{tA_f}\right\| \le M_f e^{\omega_f t}, \qquad \left\|e^{tA_c}\right\| \le M_c e^{\omega_c t}.$$

Then

$$\|\mathfrak{D}(t)\| \le M_f M_c \|\Delta\| \int_0^t e^{\omega_c(t-r)} e^{\omega_f r}\, dr.$$

For contraction semigroups,

$$\boxed{\|\mathfrak{D}(t)\| \le t \|\Delta\|.}$$

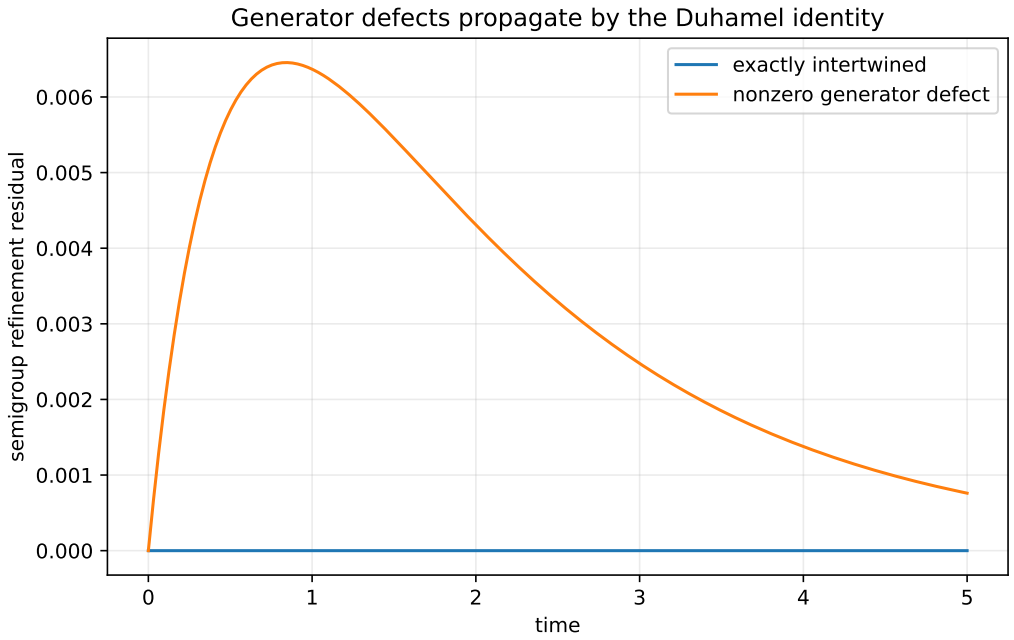


Figure 3: Exact generator intertwining gives a vanishing semigroup defect. A controlled perturbation produces a nonzero defect propagated by the Duhamel identity.

3.1 Resolvent form

For z in the common resolvent set, let

$$R_f(z) = (zI - A_f)^{-1}, \qquad R_c(z) = (zI - A_c)^{-1}.$$

Theorem 3.3 (Resolvent refinement identity)**.**

$$\boxed{CR_f(z) - R_c(z)C = R_c(z)\Delta R_f(z).}$$

Proof. Multiply the left side by $zI - A_c$ on the left and $zI - A_f$ on the right. The result is

$$(zI - A_c)C - C(zI - A_f) = CA_f - A_cC = \Delta.$$

Multiplication by the two resolvents recovers the stated identity. □

4 Markov seeds and stochastic transfer

Use column probability vectors. A finite continuous-time Markov generator Q has nonnegative off-diagonal entries and column sums zero. The stochastic semigroup is

$$K(t) = e^{tQ}.$$

Theorem 4.1 (Finite Markov semigroup)**.** *If Q has nonnegative off-diagonal entries and column sums zero, then e^{tQ} has nonnegative entries and column sums one for every $t \geq 0$.*

Proof. Choose $\lambda \geq \max_i(-Q_{ii})$. Then $B = Q + \lambda I$ has nonnegative entries, so e^{tB} is entrywise nonnegative by its power series. Hence

$$e^{tQ} = e^{-\lambda t} e^{tB}$$

is nonnegative. Since $\mathbf{1}^T Q = 0$, differentiation gives

$$\frac{d}{dt}\mathbf{1}^T e^{tQ} = \mathbf{1}^T Q e^{tQ} = 0,$$

so $\mathbf{1}^T e^{tQ} = \mathbf{1}^T$. □

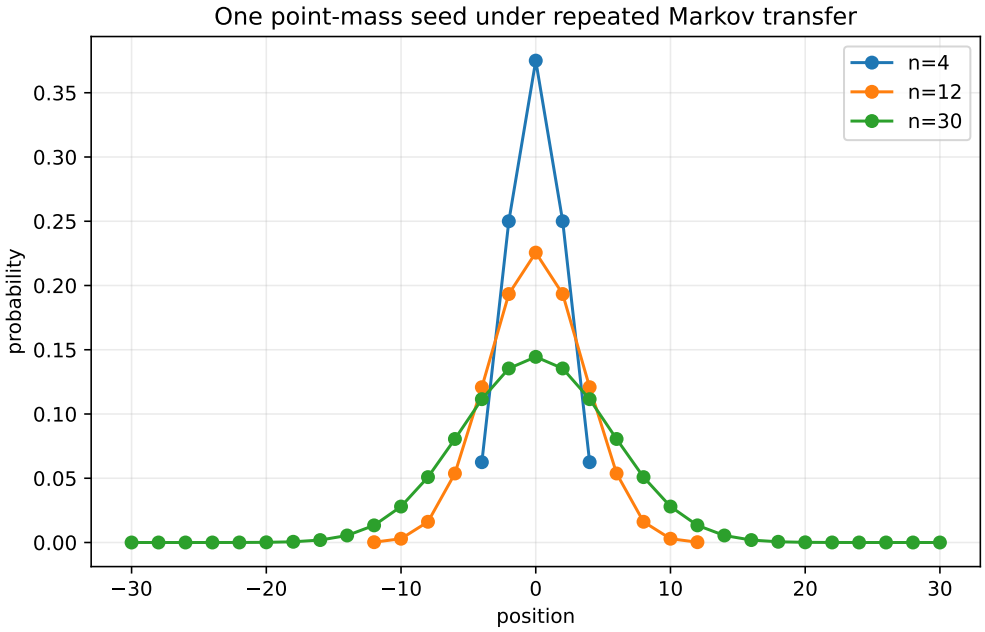


Figure 4: The point mass δ_0 propagated by repeated symmetric random-walk transfer.

Example 4.2 (Symmetric random walk)**.** Starting from δ_0 and applying the transfer

$$T = \frac{1}{2}(S + S^{-1})$$

for n steps gives

$$\mathbb{P}(X_n = k) = 2^{-n}\binom{n}{(n+k)/2}$$

when $k \equiv n \pmod 2$, and zero otherwise. Hence

$$\mathbb{E}[X_n] = 0, \qquad \mathrm{Var}(X_n) = n.$$

The parity packet is exact: all mass remains in the one residue class permitted by n.

5 Markov lumping as exact refinement

Let K be a column-stochastic fine transition matrix, let C aggregate fine states into coarse packets, and let $\bar{K}$ be a coarse transition matrix. Define

$$\Delta = CK - \bar{K}C.$$

Theorem 5.1 (Exact lumpability)**.** *If $\Delta = 0$, then*

$$\boxed{CK^n = \bar{K}^n C}$$

for every $n \geq 0$. Thus fine evolution followed by coarse reconstruction equals coarse evolution from the beginning.

Proof. The result is immediate for $n = 0$. If $CK^n = \bar{K}^n C$, then

$$CK^{n+1} = \bar{K}^n CK = \bar{K}^{n+1} C.$$

□

Theorem 5.2 (Finite-horizon defect formula)**.** *For arbitrary $\Delta = CK - \bar{K}C$,*

$$\boxed{CK^n - \bar{K}^n C = \sum_{j=0}^{n-1} \bar{K}^{n-1-j} \Delta K^j.}$$

If K and $\bar{K}$ are contractions in the chosen norm, then

$$\left\| CK^n - \bar{K}^n C \right\| \leq n \left\| \Delta \right\|.$$

Proof. The identity follows by telescoping:

$$\bar{K}^{n-1-j} CK^{j+1} - \bar{K}^{n-j} CK^j = \bar{K}^{n-1-j} (CK - \bar{K}C) K^j.$$

Summing from $j = 0$ to $n - 1$ leaves $CK^n - \bar{K}^n C$. The norm bound follows termwise. □

6 Conditional expectation as a stochastic projector

Let $\Omega = \{1, \ldots, n\}$ carry strictly positive probabilities π_i, and let $\mathcal{Q} = \{B_1, \ldots, B_m\}$ be a partition. Define

$$(P_{\mathcal{Q}} f)_i = \frac{\sum_{j \in B(i)} \pi_j f_j}{\sum_{j \in B(i)} \pi_j},$$

where $B(i)$ is the block containing i.

Theorem 6.1 (Finite conditional-expectation projector)**.** *With the weighted inner product*

$$\langle f, g \rangle_\pi = \sum_i \pi_i f_i \overline{g_i},$$

the operator $P_{\mathcal{Q}}$ satisfies

$$P_{\mathcal{Q}}^2 = P_{\mathcal{Q}}, \qquad \langle P_{\mathcal{Q}} f, g \rangle_\pi = \langle f, P_{\mathcal{Q}} g \rangle_\pi.$$

Its range is the space of functions constant on each partition block.

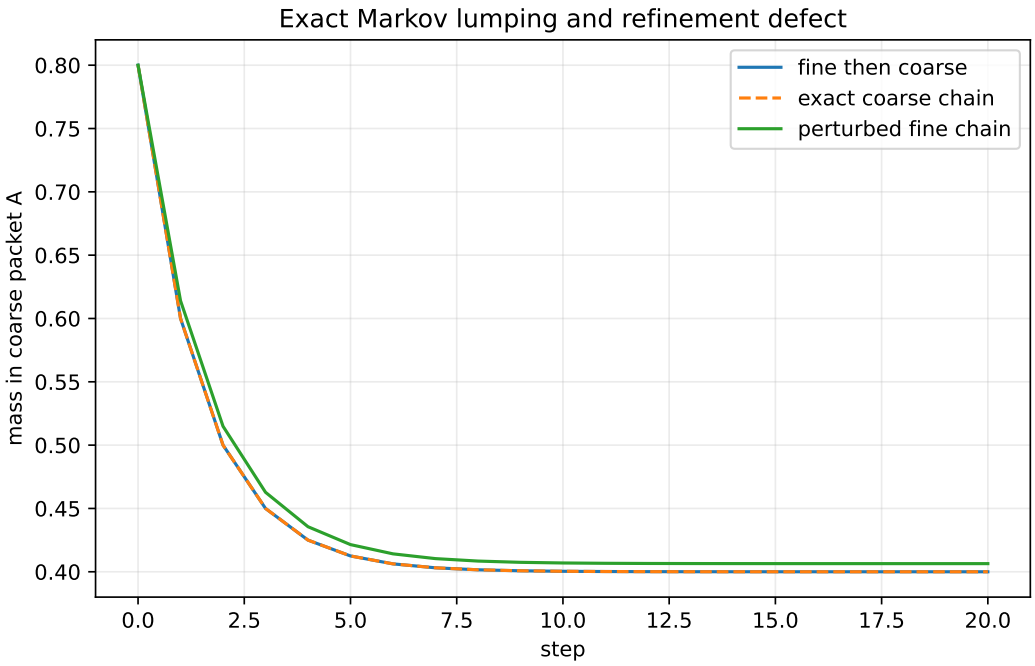


Figure 5: Exact lumping agrees at every horizon. A small generator-level or one-step defect creates a controlled coarse discrepancy.

Proof. After one application, the result is constant on every block, so applying the same block average again changes nothing. Weighted self-adjointness follows by expanding both inner products block by block; both equal

$$\sum_{B\in\mathcal{Q}} \frac{1}{\pi(B)} \left(\sum_{i\in B} \pi_i f_i\right) \overline{\left(\sum_{j\in B} \pi_j g_j\right)}.$$

□

Theorem 6.2 (Tower law and martingale packets)**.** *If $\mathcal{Q}_c$ is coarser than $\mathcal{Q}_f$, then*

$$\boxed{P_c P_f = P_f P_c = P_c.}$$

For a nested tower $P_0 \le P_1 \le \cdots \le P_r$, define

$$D_0 = P_0, \qquad D_j = P_j - P_{j-1}.$$

Then

$$D_i D_j = \delta_{ij} D_j, \qquad P_r = \sum_{j=0}^{r} D_j.$$

Proof. A coarse-block average of a fine-block average is the coarse-block average, and a fine-block average of a coarse-constant function leaves it unchanged. The difference identities follow from $P_i P_j = P_{\min(i,j)}$. □

Classical Input 6.3 (Gaussian chaos realization). For the standard Gaussian measure, the normalized probabilists' Hermite polynomials form orthonormal chaos packets. The Ornstein–Uhlenbeck semigroup acts diagonally:

$$T_t H_n = e^{-nt} H_n.$$

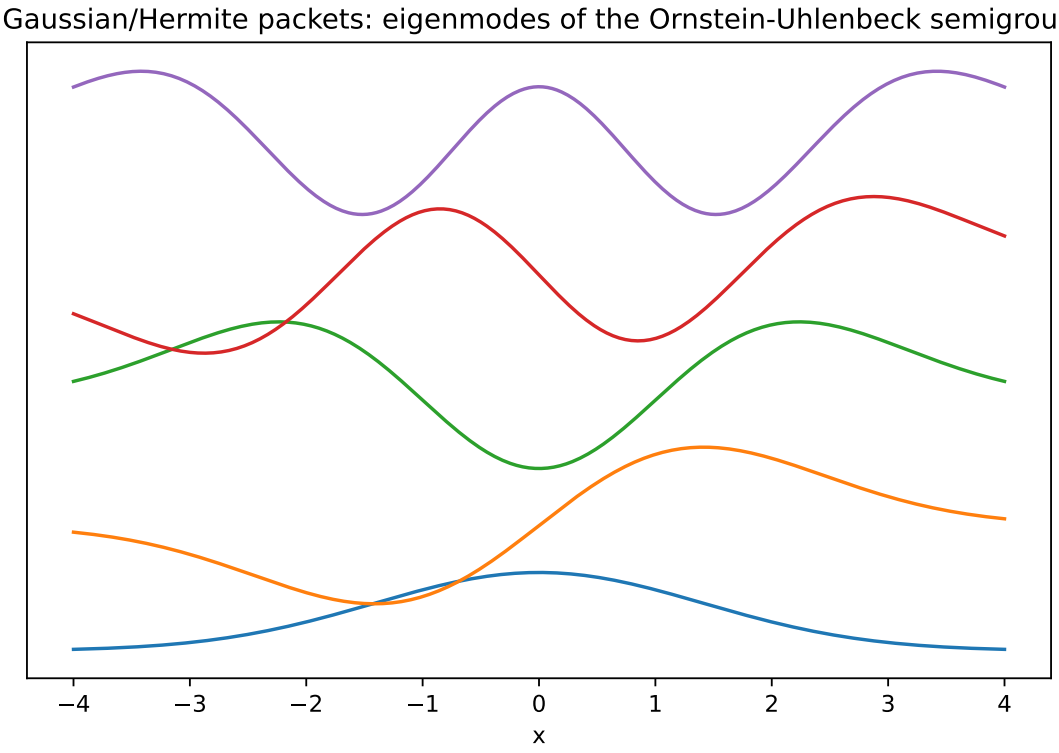


Figure 6: Hermite–Gaussian packets, which diagonalize the Ornstein–Uhlenbeck semigroup. The finite numerical release verifies their Gaussian orthogonality through quadrature.

Thus conditional expectation, martingale differences, and Gaussian chaos are all projector realizations of stochastic refinement.

7 Bayesian conditioning as a normalized nonlinear operator

Let $p = (p_i)$ be a probability vector and let $\lambda = (\lambda_i)$ be a nonnegative likelihood with positive evidence

$$Z_\lambda(p) = \sum_i \lambda_i p_i.$$

Define

$$\boxed{\mathcal{B}_\lambda(p) = \frac{\lambda \odot p}{Z_\lambda(p)}.}$$

This map is nonlinear solely because of its scalar reconstruction factor.

Theorem 7.1 (Independent-likelihood composition)**.** *For nonnegative likelihoods λ and μ, assume $Z_\lambda(p) > 0$ and $Z_{\lambda\odot\mu}(p) > 0$. Equivalently, both successive updates have positive evidence. Then*

$$\boxed{\mathcal{B}_\mu(\mathcal{B}_\lambda(p)) = \mathcal{B}_{\lambda\odot\mu}(p).}$$

Proof. The first normalization appears in both numerator and denominator of the second update and cancels:

$$\frac{\mu_i(\lambda_i p_i/Z_\lambda)}{\sum_j \mu_j\lambda_j p_j/Z_\lambda} = \frac{\mu_i\lambda_i p_i}{\sum_j \mu_j\lambda_j p_j}.$$

□

7.1 Packet evidence and reconstruction

Let the state indices be partitioned into packets I_α. Define

$$Z_\alpha = \sum_{i \in I_\alpha} \lambda_i p_i, \qquad m_\alpha = \frac{Z_\alpha}{\sum_\beta Z_\beta}, \qquad q_{\alpha,i} = \begin{cases} \lambda_i p_i / Z_\alpha, & Z_\alpha > 0,\ i \in I_\alpha, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 7.2 (Bayesian packet reconstruction)**.**

$$\boxed{\mathcal{B}_\lambda(p) = \sum_\alpha m_\alpha q_\alpha.}$$

Only packets with positive evidence carry a conditional probability vector. A zero-evidence packet has zero posterior mass and contributes the zero vector. The nonlinear coupling between packets occurs only through the total evidence used to normalize the packet evidences.

Proof. If $Z_\alpha = 0$, nonnegativity implies $\lambda_i p_i = 0$ for every $i \in I_\alpha$, so the reconstruction identity holds with both sides zero. For $Z_\alpha > 0$ and $i \in I_\alpha$,

$$m_\alpha q_{\alpha,i} = \frac{Z_\alpha}{Z} \frac{\lambda_i p_i}{Z_\alpha} = \frac{\lambda_i p_i}{Z}.$$

The packet supports are disjoint, so summing reconstructs the full posterior. □

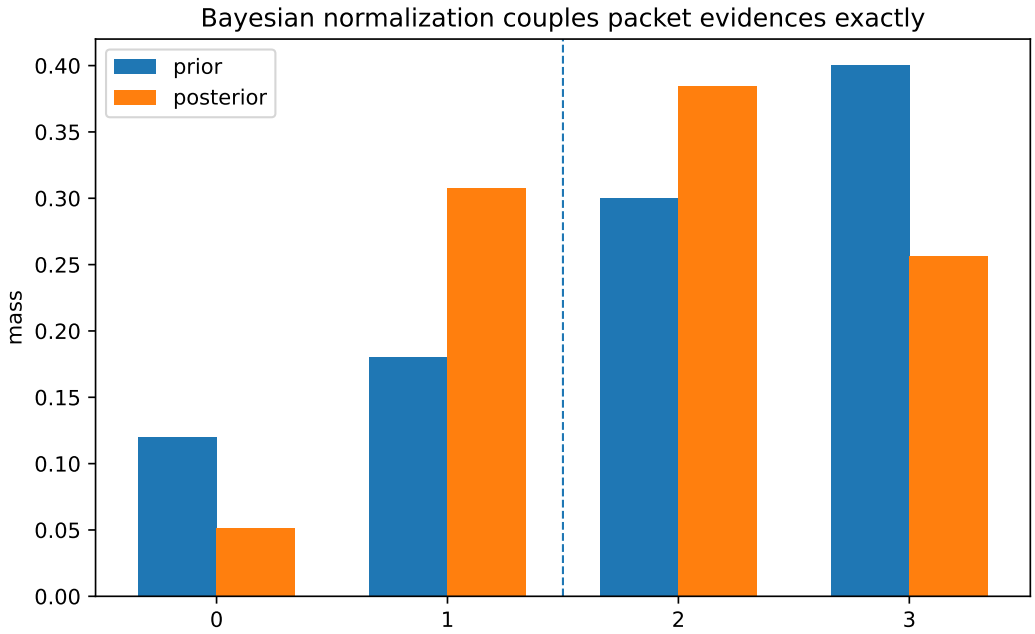


Figure 7: A finite prior and posterior split into two state packets. Evidence changes both the local packet shape and the global packet mass.

Theorem 7.3 (Likelihood perturbation bound)**.** *Let p be fixed and let $\lambda, \mu \geq 0$ have positive evidences Z_λ, Z_μ. Then*

$$\boxed{\|\mathcal{B}_\lambda(p) - \mathcal{B}_\mu(p)\|_1 \leq \frac{2 \sum_i |\lambda_i - \mu_i| p_i}{\min(Z_\lambda, Z_\mu)}.}$$

Proof. Add and subtract $\mu \odot p/Z_\lambda$ and use the triangle inequality:

$$\left\| \frac{\lambda \odot p}{Z_\lambda} - \frac{\mu \odot p}{Z_\mu} \right\|_1 \le \frac{\|(\lambda - \mu) \odot p\|_1}{Z_\lambda} + \|\mu \odot p\|_1 \left| \frac{1}{Z_\lambda} - \frac{1}{Z_\mu} \right|.$$

The second term equals $|Z_\lambda - Z_\mu|/Z_\lambda$ and is bounded by the same weighted L^1 difference divided by Z_λ. Interchanging λ and μ gives the minimum-evidence form. □

8 Exact Bayesian refinement and information-loss packets

Let the fine state set I be partitioned into nonempty coarse packets I_a, $a \in A$. The coarsening map on probability vectors is

$$(C_\pi p)_a = P_a = \sum_{i \in I_a} p_i.$$

For a fine likelihood $\lambda = (\lambda_i)_{i \in I}$ and a prior p, define the effective coarse likelihood

$$r_a(p) = \frac{\sum_{i \in I_a} \lambda_i p_i}{P_a} \qquad (P_a > 0).$$

Its value on a zero-mass packet is irrelevant and may be set to zero.

Theorem 8.1 (Exact effective-likelihood factorization)**.** *Whenever the fine evidence is positive,*

$$\boxed{C_\pi \mathcal{B}_\lambda(p) = \mathcal{B}_{r(p)}(C_\pi p).}$$

Thus the only fine information needed for the coarse posterior, beyond the coarse prior masses, is the vector of conditional packet evidences $r(p)$.

Proof. For $P = C_\pi p$ and $Z = \sum_i \lambda_i p_i = \sum_a P_a r_a(p)$,

$$\big(C_\pi \mathcal{B}_\lambda(p)\big)_a = \frac{\sum_{i \in I_a} \lambda_i p_i}{Z} = \frac{P_a r_a(p)}{\sum_b P_b r_b(p)} = \big(\mathcal{B}_{r(p)}(P)\big)_a.$$

□

The next result gives the exact necessary and sufficient condition for a coarse likelihood to work for every prior.

Theorem 8.2 (Bayesian refinement criterion)**.** *Assume that the partition contains at least two coarse packets and that the fine and coarse likelihoods are strictly positive. The identity*

$$\boxed{C_\pi \mathcal{B}_\lambda(p) = \mathcal{B}_{\bar{\lambda}}(C_\pi p)}$$

holds for every fine prior p if and only if there exists one scalar $c > 0$ such that

$$\boxed{\lambda_i = c\, \bar{\lambda}_a \qquad (i \in I_a).}$$

Equivalently, the likelihood is constant on each coarse packet, with the coarse values determined up to the globally invisible Bayesian scaling c.

Proof. If $\lambda_i = c\bar{\lambda}_a$ on I_a, then $P_a r_a(p) = c\bar{\lambda}_a P_a$ for every prior, including zero-mass packets, and the preceding theorem gives the desired identity because Bayesian normalization removes the common scalar c.

Conversely, fix two fine states $i, i' \in I_a$ and one state $j \in I_b$ in another coarse packet. Vary the distribution of a fixed coarse mass inside I_a between i and i' while leaving every coarse packet mass unchanged. The right side depends only on those coarse masses and is therefore unchanged. The left-side posterior mass of packet a depends on the conditional average of λ_i and $\lambda_{i'}$. It can be independent of that variation only if $\lambda_i = \lambda_{i'}$. Hence λ is constant on every packet; write the value on I_a as λ_a.

Now use priors supported on one state of packet a and one state of packet b. Equality of the two coarse posterior odds gives

$$\frac{\lambda_a}{\lambda_b} = \frac{\bar{\lambda}_a}{\bar{\lambda}_b}.$$

This holds for every pair of coarse packets, so $\lambda_a = c\bar{\lambda}_a$ for one common $c > 0$. □

Remark 8.3 (The one-packet boundary). If the coarse partition has only one packet, every posterior coarsens to the scalar 1, so commutation is automatic and no within-packet likelihood information can be identified. The nontrivial criterion begins with at least two coarse packets.

When sufficiency fails, the defect still has an exact finite factorization. Fix a declared coarse likelihood $\bar{\lambda}$, put

$$P = C_\pi p, \qquad \delta(p) = r(p) - \bar{\lambda}, \qquad \bar{b} = \mathcal{B}_{\bar{\lambda}}(P), \qquad Z_r = \langle r(p), P \rangle.$$

Theorem 8.4 (Bayesian information-loss packet)**.** *Assume nonnegative fine and declared coarse likelihoods, with both $Z_r > 0$ and $Z_{\bar{\lambda}} > 0$. The coarsening defect is*

$$\boxed{C_\pi \mathcal{B}_\lambda(p) - \mathcal{B}_{\bar{\lambda}}(P) = \frac{1}{Z_r}\left(\operatorname{diag}(P) - \bar{b}P^{\mathsf{T}}\right)\delta(p).}$$

The factor matrix annihilates $\bar{\lambda}$. This records the common-scale null direction at the declared coarse likelihood; the full exact defect remains homogeneous under simultaneous likelihood rescaling because the numerator and evidence denominator scale together. Moreover,

$$\boxed{\left\|C_\pi \mathcal{B}_\lambda(p) - \mathcal{B}_{\bar{\lambda}}(P)\right\|_1 \le \frac{2\sum_a P_a |\delta_a(p)|}{\min\{Z_r, Z_{\bar{\lambda}}\}}}$$

and

$$\sum_a P_a |\delta_a(p)| \le \sum_i p_i \left|\lambda_i - \bar{\lambda}_{\pi(i)}\right|.$$

Proof. The effective-likelihood theorem reduces the defect to

$$\mathcal{B}_{\bar{\lambda}+\delta}(P) - \mathcal{B}_{\bar{\lambda}}(P).$$

Since $Z_r = Z_{\bar{\lambda}} + P^{\mathsf{T}}\delta$, direct subtraction gives

$$\frac{1}{Z_r}\left(\operatorname{diag}(P) - \bar{b}P^{\mathsf{T}}\right)\delta.$$

Also,

$$\left(\operatorname{diag}(P) - \bar{b}P^{\mathsf{T}}\right)\bar{\lambda} = P \odot \bar{\lambda} - \bar{b}Z_{\bar{\lambda}} = 0.$$

The first norm bound is the likelihood perturbation theorem applied on the coarse simplex. For a zero-mass packet, both sides of the following estimate vanish. For $P_a > 0$,

$$P_a|\delta_a| = P_a \left| \sum_{i \in I_a} \frac{p_i}{P_a} (\lambda_i - \bar{\lambda}_a) \right| \leq \sum_{i \in I_a} p_i |\lambda_i - \bar{\lambda}_a|,$$

and summing over a proves the second inequality. □

Corollary 8.5 (Nested Bayesian packet tower)**.** *Consider a nested finite packet tower with strictly positive likelihoods at every level. At each adjacent coarsening with at least two coarse packets, commutation for every prior is equivalent to the fine likelihood being blockwise proportional to the declared coarse likelihood, with one common positive proportionality factor. An adjacent coarsening to a single packet commutes automatically and imposes no such condition. Under these conditions, all composite coarsening squares commute.*

Proof. At each adjacent coarsening with at least two coarse packets, apply the refinement criterion. For a single coarse packet, both normalized posteriors equal 1. Composition of these commuting squares gives the full tower statement. □

Corollary 8.6 (Canonical coarsest likelihood-resolving partition)**.** *For a finite family of likelihoods* $\{\lambda^{(y)}\}_{y \in Y}$*, define*

$$i \sim j \quad \Longleftrightarrow \quad \lambda_i^{(y)} = \lambda_j^{(y)} \text{ for every } y \in Y.$$

The equivalence classes form the unique coarsest likelihood-resolving partition: every likelihood is constant on each packet. Every update therefore closes on this partition whenever its evidence is positive. If all likelihoods are strictly positive and there are at least two profile classes, this is also the unique coarsest nontrivial partition on which every update closes for every prior. The one-packet partition always closes, independently of the profiles, and is excluded from this last assertion.

Proof. Equality of likelihood profiles is an equivalence relation. Its classes are likelihood-resolving by definition, and every likelihood-resolving partition refines them. Exact closure follows by summing the weighted masses in each fiber, including zero-mass fibers. Under strict positivity, the refinement criterion implies that every nontrivial all-prior closed partition is likelihood-resolving. When every profile is identical, all partitions close; there need not be a unique coarsest nontrivial partition. □

Numerical Certificate 8.7 (Bayesian refinement). For the finite test partition $\{0,1\} \sqcup \{2,3\}$, a blockwise-constant likelihood gave commutation residual 1.11×10^{-16}. A nonconstant likelihood produced an exact defect-factorization residual 7.29×10^{-17}. Its measured L^1 defect was 4.8787×10^{-2}, below the certified bound 1.1016×10^{-1}. Two priors with identical coarse masses but different hidden within-packet shapes produced coarse posteriors differing by 1.2400×10^{-1}, which is a direct necessity witness.

Boundary 8.8 (Finite exact theorem). The conjectured Bayesian refinement law is therefore resolved for finite packet systems and arbitrary priors. Infinite-dimensional disintegration, posterior consistency, and stability of data-dependent partitions require their own measure-theoretic and asymptotic hypotheses.

9 Bayesian flows, conjugacy, and hidden states

Fix a score vector g. The likelihood flow is

$$\lambda_i(t) = e^{t g_i},$$

and the Bayesian flow is

$$\Phi_t(p)_i = \frac{e^{t g_i} p_i}{\sum_j e^{t g_j} p_j}.$$

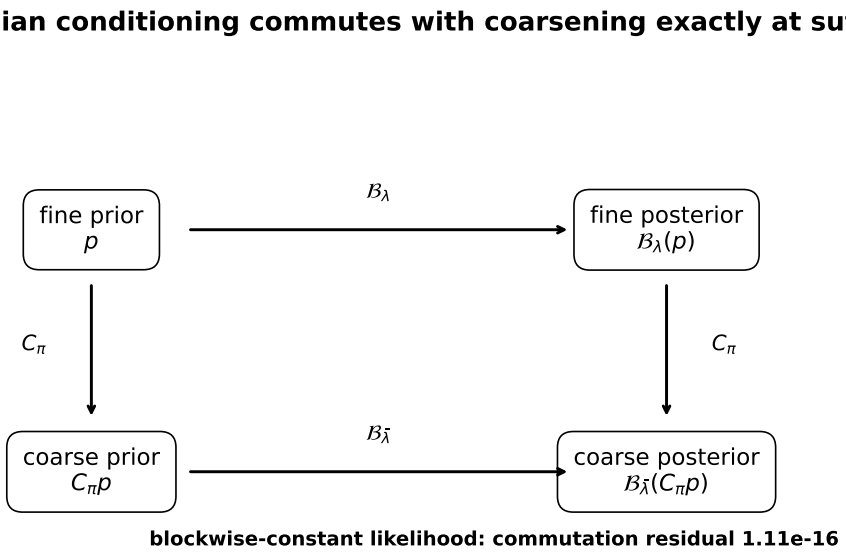


Figure 8: For strictly positive likelihoods and at least two coarse packets, the Bayesian refinement square commutes for every prior exactly when the likelihood is constant on each coarsening fiber, up to one global scale.

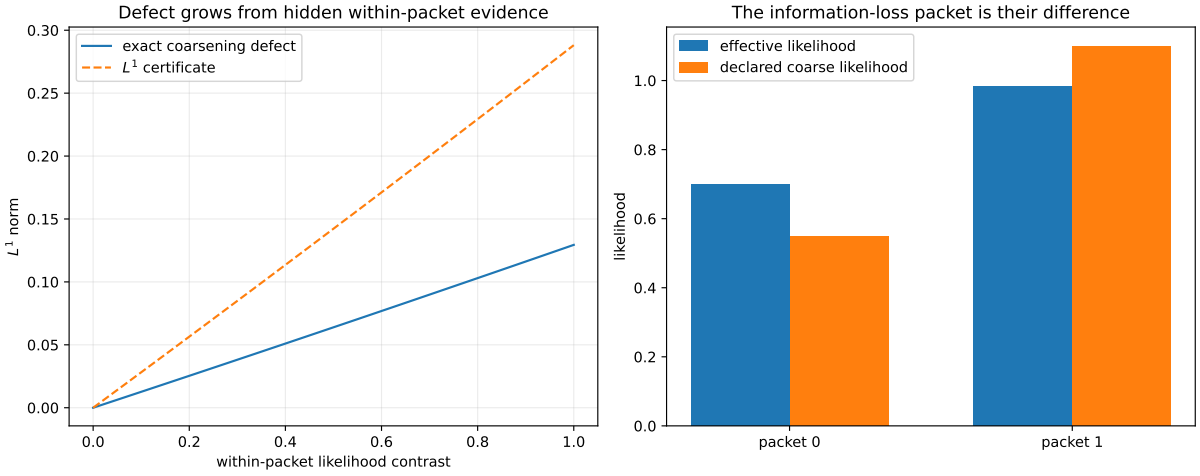


Figure 9: When packet sufficiency fails, the exact defect factors through the difference between the conditional effective likelihood and the declared coarse likelihood.

Theorem 9.1 (Bayesian nonlinear semigroup)**.**

$$\boxed{\Phi_{s+t} = \Phi_s \circ \Phi_t.}$$

Its generator is the centered multiplicative field

$$\boxed{\frac{d}{dt}p_i(t) = (g_i - \langle g, p(t)\rangle)p_i(t).}$$

Proof. The semigroup law is the likelihood-composition theorem with $e^{sg}e^{tg} = e^{(s+t)g}$. Differentiating the normalized exponential formula gives

$$\dot{p}_i = g_i p_i - p_i \frac{\sum_j g_j e^{tg_j} p_j(0)}{\sum_j e^{tg_j} p_j(0)}.$$

The quotient is $\langle g, p(t)\rangle$. □

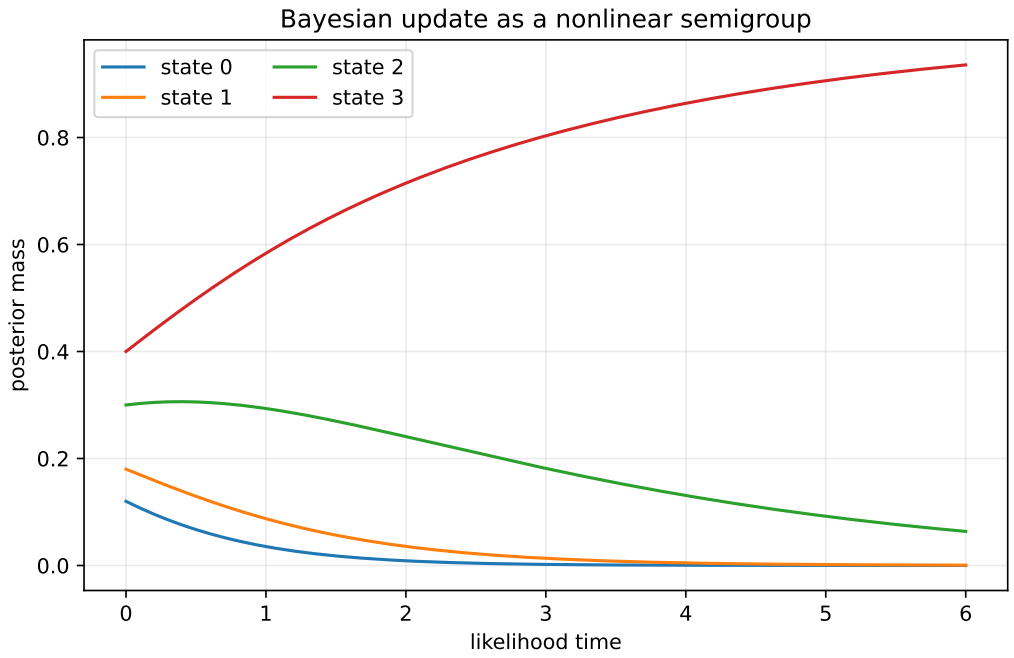


Figure 10: Continuous likelihood accumulation produces a nonlinear semigroup on the probability simplex.

Example 9.2 (Beta–Binomial conjugate packet)**.** A Beta(a, b) prior and s successes in n trials produce

$$\text{Beta}(a + s, b + n - s).$$

The prior exponents are additive packets: the likelihood increments them by $(s, n-s)$. In the numerical example,

$$(a, b) = (2, 3), \quad (s, n) = (7, 10),$$

so the posterior is Beta$(9, 6)$ with mean 0.6.

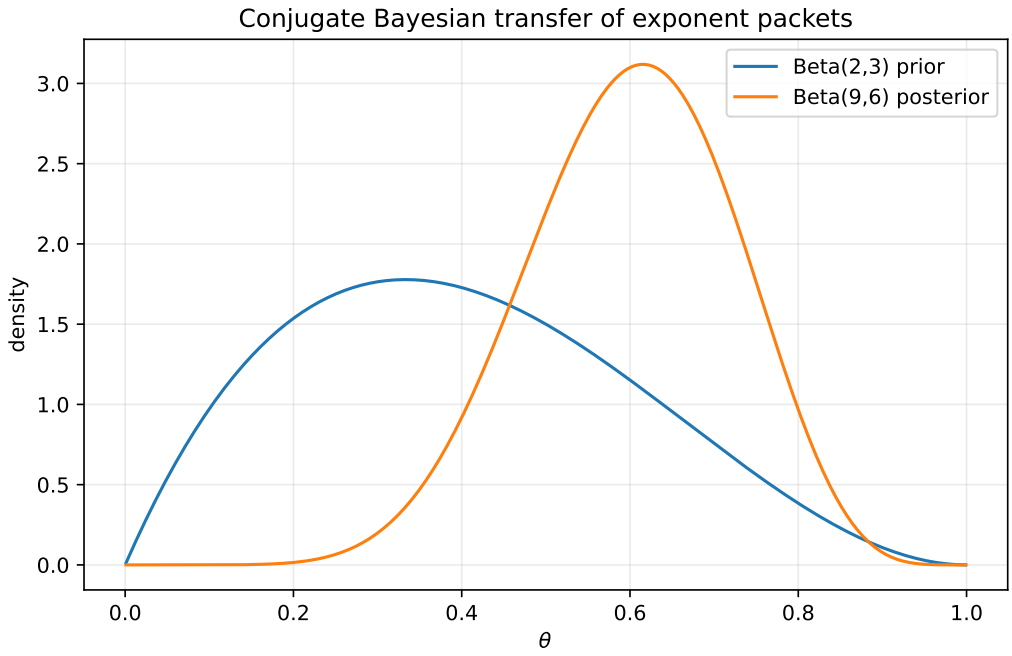


Figure 11: A conjugate Bayesian update viewed as exact transfer of exponent packets.

9.1 Hidden Markov filtering

Let K be column stochastic and let $\lambda^{(r)}$ be the likelihood associated with the rth observation. The filter recursion is

$$\widehat{p}_r = K p_{r-1}, \qquad p_r = \mathcal{B}_{\lambda^{(r)}}(\widehat{p}_r).$$

Proposition 9.3 (Prediction-update reconstruction)**.** *Every hidden Markov filtering step is the composition of a linear Markov transfer and a normalized nonlinear likelihood transfer. Assume a probability initial vector, a column-stochastic transition matrix, and nonnegative likelihoods with positive evidence at every step. Then every p_r is a probability vector.*

Proof. Column stochasticity gives $\widehat{p}_r \geq 0$ and $\mathbf{1}^T \widehat{p}_r = 1$. Bayesian normalization then gives $p_r \geq 0$ and $\mathbf{1}^T p_r = 1$. □

10 Fractional powers as spectral packet transfer

Let $L = L^* \geq 0$ on a finite-dimensional Hilbert space. Its spectral resolution is

$$L = \sum_j \lambda_j P_j.$$

For $\alpha > 0$, define

$$\boxed{L^\alpha = \sum_j \lambda_j^\alpha P_j.}$$

The associated fractional semigroup is

$$T_\alpha(t) = e^{-tL^\alpha} = \sum_j e^{-t\lambda_j^\alpha} P_j.$$

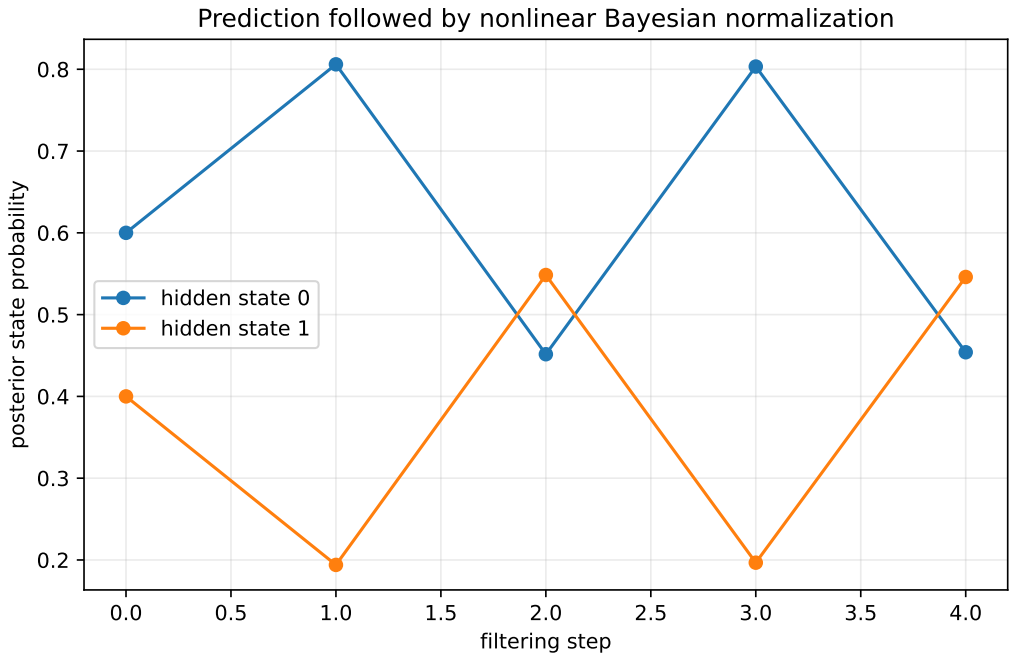


Figure 12: A hidden Markov filter alternates a linear prediction semigroup with nonlinear Bayesian normalization.

Theorem 10.1 (Fractional packet calculus)**.** *For $\alpha, \beta > 0$,*

$$L^{\alpha} L^{\beta} = L^{\alpha+\beta},$$

and for $s, t \geq 0$,

$$\boxed{T_{\alpha}(s) T_{\alpha}(t) = T_{\alpha}(s+t).}$$

Every spectral packet P_j reduces L^{α} and $T_{\alpha}(t)$.

Proof. All statements hold coefficientwise on the mutually orthogonal spectral projectors. For example,

$$L^{\alpha} L^{\beta} = \sum_{j,k} \lambda_j^{\alpha} \lambda_k^{\beta} P_j P_k = \sum_{j} \lambda_j^{\alpha+\beta} P_j.$$

□

10.1 Square-root subordination

Theorem 10.2 (Poisson subordination)**.** *For $t > 0$ and $L \geq 0$,*

$$\boxed{e^{-t\sqrt{L}} = \frac{t}{2\sqrt{\pi}} \int_0^{\infty} s^{-3/2} e^{-t^2/(4s)} e^{-sL} \, ds.}$$

The integral converges in operator norm in finite dimension.

Proof. By the spectral theorem it suffices to prove the scalar identity

$$e^{-t\sqrt{\lambda}} = \frac{t}{2\sqrt{\pi}} \int_0^{\infty} s^{-3/2} e^{-t^2/(4s) - \lambda s} \, ds$$

for $\lambda \geq 0$. This is the classical Gaussian integral after the substitution $u = t/(2\sqrt{s})$. Applying the identity to every eigenvalue and recombining the spectral projectors proves the operator formula. □

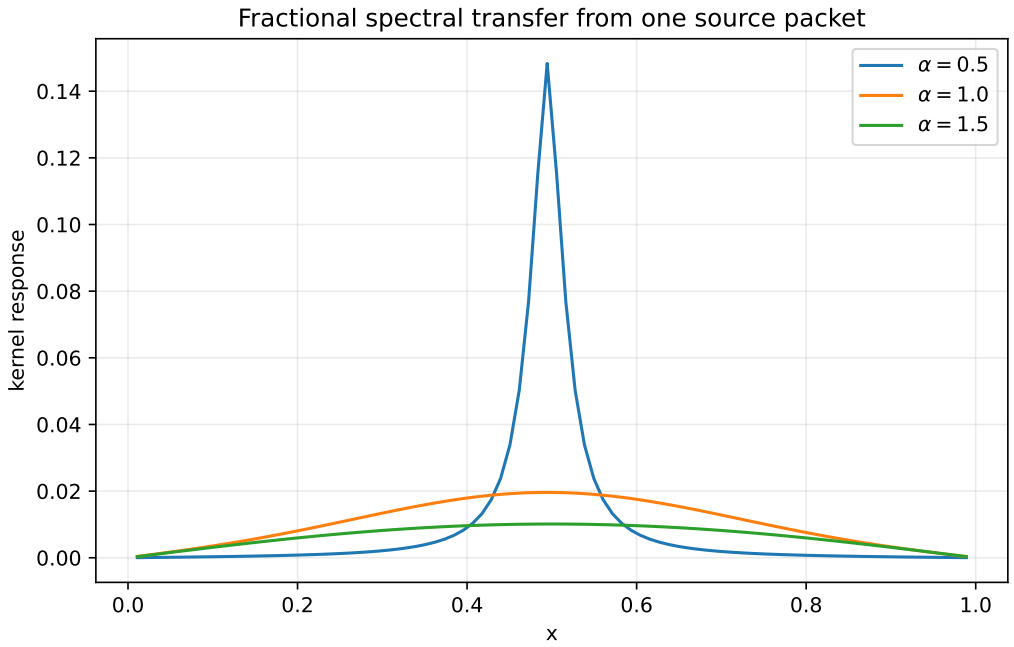


Figure 13: Different spectral exponents produce different transfer profiles from the same point seed.

10.2 Fractional naturality under a bounded refinement defect

Let $A \geq 0$ act on a fine Hilbert space, let $B \geq 0$ act on a coarse Hilbert space, and let C be a bounded coarsening map. Set

$$D = CA - BC, \qquad D_\alpha = CA^\alpha - B^\alpha C, \qquad 0 < \alpha < 1.$$

Theorem 10.3 (Fractional intertwining theorem)**.** *For bounded nonnegative self-adjoint A and B,*

$$\boxed{D_\alpha = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty t^\alpha (tI + B)^{-1} D (tI + A)^{-1}\, dt.}$$

The improper integral converges in operator norm. Moreover,

$$\boxed{\|D_\alpha\| \leq K_\alpha \|C\|^{1-\alpha} \|D\|^\alpha, \qquad K_\alpha = \frac{2^{1-\alpha} \sin(\pi\alpha)}{\pi\alpha(1-\alpha)}.}$$

If $A, B \geq mI$ for some $m > 0$, then the stronger linear estimate holds:

$$\boxed{\|D_\alpha\| \leq \alpha m^{\alpha-1} \|D\|.}$$

In particular, exact generator intertwining $CA = BC$ implies exact fractional intertwining $CA^\alpha = B^\alpha C$.

Proof. For $0 < \alpha < 1$, the Balakrishnan formula gives

$$A^\alpha = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty t^{\alpha-1} A (tI + A)^{-1}\, dt,$$

and likewise for B. Using

$$A(tI + A)^{-1} = I - t(tI + A)^{-1}$$

and the rectangular resolvent identity

$$(tI + B)^{-1}C - C(tI + A)^{-1} = (tI + B)^{-1}(CA - BC)(tI + A)^{-1},$$

the difference of the two Balakrishnan integrals becomes the displayed resolvent-defect formula.

For small t,

$$\left\|CA(tI + A)^{-1} - B(tI + B)^{-1}C\right\| \le 2\|C\|,$$

because the two positive resolvent factors are contractions. For large t, the resolvent identity gives the bound $\|D\|/t$. Splitting the integral at

$$r = \frac{\|D\|}{2\|C\|}$$

when C and D are nonzero yields

$$\begin{aligned}\|D_\alpha\| &\le \frac{\sin(\pi\alpha)}{\pi}\left(2\|C\|\int_0^r t^{\alpha-1}\,dt + \|D\|\int_r^\infty t^{\alpha-2}\,dt\right)\\ &= K_\alpha\|C\|^{1-\alpha}\|D\|^{\alpha}.\end{aligned}$$

The zero cases are immediate.

If $A, B \ge mI$, then both resolvents have norm at most $(t + m)^{-1}$, so

$$\|D_\alpha\| \le \frac{\sin(\pi\alpha)}{\pi}\|D\|\int_0^\infty \frac{t^\alpha}{(t+m)^2}\,dt.$$

After $t = mu$, the beta integral equals

$$m^{\alpha-1}\frac{\alpha\pi}{\sin(\pi\alpha)},$$

which gives the linear estimate. □

Corollary 10.4 (Fractional semigroup refinement)**.** *For $s \ge 0$,*

$$\boxed{Ce^{-sA^\alpha} - e^{-sB^\alpha}C = -\int_0^s e^{-(s-r)B^\alpha}D_\alpha e^{-rA^\alpha}\,dr.}$$

Hence

$$\left\|Ce^{-sA^\alpha} - e^{-sB^\alpha}C\right\| \le s\|D_\alpha\|.$$

Under the common gap $A, B \ge mI$,

$$\boxed{\left\|Ce^{-sA^\alpha} - e^{-sB^\alpha}C\right\| \le se^{-sm^\alpha}\alpha m^{\alpha-1}\|D\|.}$$

Proof. Differentiate

$$r \longmapsto e^{-(s-r)B^\alpha}Ce^{-rA^\alpha}$$

and integrate from 0 to s. Positive fractional semigroups are contractions; under the gap, the product of their two norm bounds is e^{-sm^α}. □

Corollary 10.5 (Fractional refinement towers)**.** *Let $D_\ell = C_\ell A_{\ell+1} - A_\ell C_\ell$ in a bounded nonnegative self-adjoint tower, with $\sup_\ell \|C_\ell\| \le M$. If*

$$\sum_\ell \|D_\ell\|^\alpha < \infty,$$

then the fractional defects

$$C_\ell A_{\ell+1}^\alpha - A_\ell^\alpha C_\ell$$

are summable. If the tower has a uniform gap $m > 0$, it is enough that $\sum_\ell \|D_\ell\| < \infty$. On every bounded time interval, the corresponding fractional-semigroup defects are summable as well.

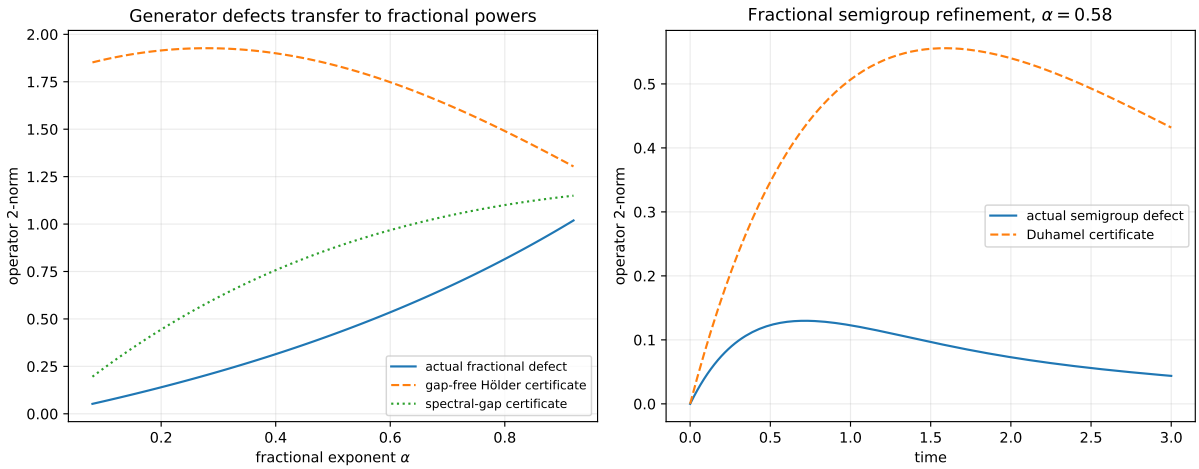


Figure 14: A generator-level refinement defect transfers quantitatively to fractional powers and then to fractional semigroups. The two upper curves are theorem certificates, not fitted error bars.

Boundary 10.6 (Unbounded and nonnormal generators). The theorem resolves fractional naturality for bounded nonnegative self-adjoint refinement systems and gives the exact estimate needed for finite discretization towers. For unbounded positive generators, the same resolvent argument requires a declared common-domain condition and a bounded extension of $CA - BC$. Sectorial or nonnormal generators require their own contour calculus and resolvent bounds. Those domain questions are not inferred from the finite certificates.

Boundary 10.7 (Non-self-adjoint fractional powers). The spectral definition above is complete for finite self-adjoint positive operators. Sectorial and nonnormal operators require a resolvent or contour functional calculus; the same packet language can be used only after those analytic hypotheses are established.

11 Green operators and the Poisson seed

On the interval $(0, 1)$ with Dirichlet boundary conditions, the Green kernel for $-u'' = f$ is

$$\boxed{G(x, y) = \min(x, y) - xy.}$$

Thus the point source δ_y generates the solution kernel, and a general source is reconstructed by

$$u(x) = \int_0^1 G(x, y) f(y)\, dy.$$

For n interior points with spacing $h = 1/(n + 1)$, define

$$L_h = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{pmatrix}.$$

Theorem 11.1 (Exact discrete Green reconstruction at grid nodes). *Let $x_i = ih$. Then*

$$\boxed{\frac{1}{h}(L_h^{-1})_{ij} = G(x_i, x_j).}$$

Thus the scaled discrete inverse reproduces the continuum Dirichlet Green kernel exactly at the grid nodes.

Proof. The inverse of the unscaled tridiagonal matrix has entries

$$\frac{\min(i,j)(n+1-\max(i,j))}{n+1}.$$

Multiplying by h^2 and then dividing by h gives

$$h\min(i,j)\left(1-\frac{\max(i,j)}{n+1}\right)=\min(x_i,x_j)-x_ix_j.$$

□



Figure 15: A section of the continuum Green kernel and the scaled discrete inverse. They agree at the grid nodes.

11.1 Low-mode convergence

The kth discrete eigenvalue is

$$\lambda_{k,h}=\frac{4}{h^2}\sin^2\left(\frac{k\pi h}{2}\right),$$

whereas the continuum eigenvalue is $(k\pi)^2$.

Theorem 11.2 (Second-order low-mode convergence)**.** *For fixed k,*

$$\lambda_{k,h}=(k\pi)^2+O(h^2).$$

More precisely, for sufficiently small h,

$$0\le(k\pi)^2-\lambda_{k,h}\le\frac{(k\pi)^4}{12}h^2.$$

Proof. For $x\ge0$, $0\le x^2-\sin^2x\le x^4/3$. Set $x=k\pi h/2$ and multiply by $4/h^2$. □

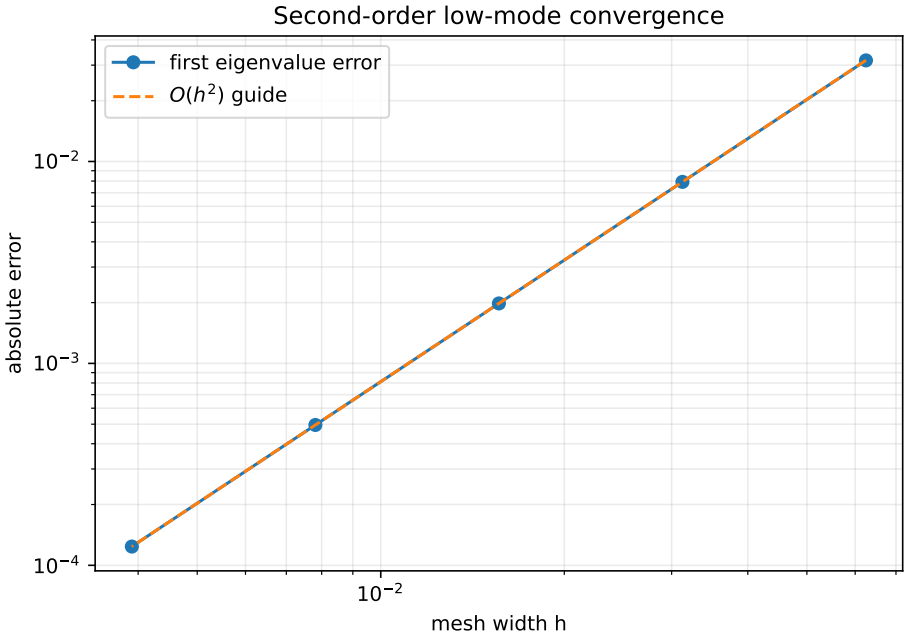


Figure 16: The first finite-difference eigenvalue converges with observed order approximately two.

12 Finite-state Feynman–Kac transfer

Let K be a column-stochastic matrix, let V be a potential on the states, and let f be terminal data. Define

$$M_V = \operatorname{diag}(e^{-V})K^T.$$

The backward Feynman–Kac value after n steps is

$$u_n = M_V^n f.$$

Theorem 12.1 (Discrete Feynman–Kac path formula)**.** *For an initial state i,*

$$\boxed{(u_n)_i = \mathbb{E}_i\left[\exp\left(-\sum_{r=0}^{n-1} V(X_r)\right) f(X_n)\right].}$$

Proof. For $n = 1$,

$$(M_V f)_i = e^{-V(i)} \sum_j K_{ji} f_j,$$

which is the stated expectation. Iterating and expanding the matrix product sums over every path $(i = i_0, i_1, \ldots, i_n)$ with probability $\prod_r K_{i_{r+1},i_r}$ and weight $\prod_r e^{-V(i_r)}$. □

13 Exact elimination of latent packets

Let $X = PX \oplus QX$ and write

$$A = \begin{pmatrix} A_{PP} & A_{PQ} \\ A_{QP} & A_{QQ} \end{pmatrix}.$$

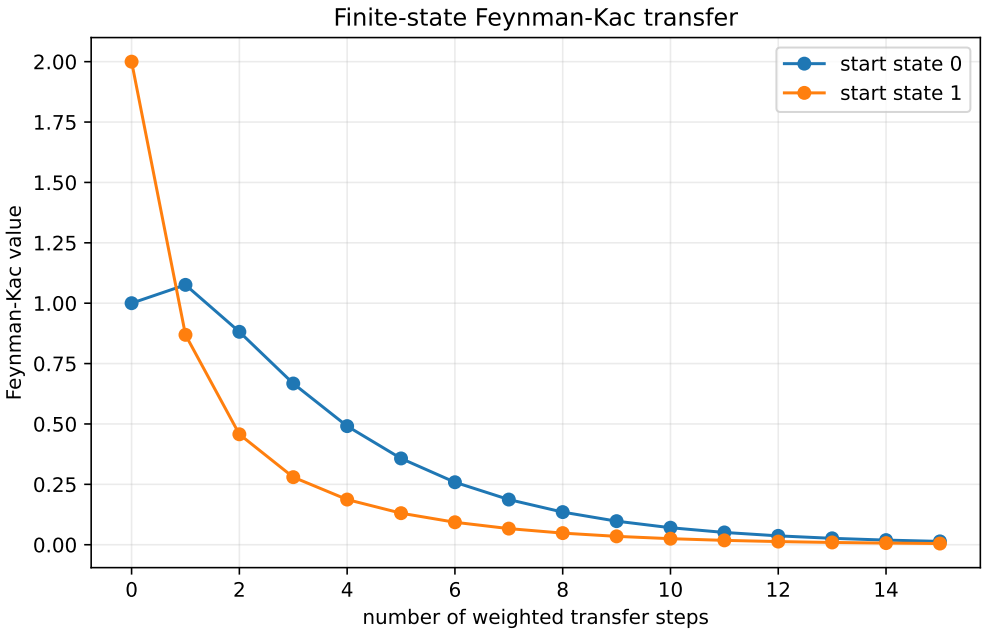


Figure 17: Potential-weighted transfer values computed by the finite Feynman–Kac operator. Exhaustive path enumeration agrees with the matrix formula in the supplied tests.

Theorem 13.1 (Schur–Feshbach projected resolvent)**.** *Assume both $zI - A$ and $zI - A_{QQ}$ are invertible. Define*

$$\boxed{F_P(z) = zI - A_{PP} - A_{PQ}(zI - A_{QQ})^{-1}A_{QP}.}$$

Then

$$\boxed{P(zI - A)^{-1}\big|_{\operatorname{Ran} P} = F_P(z)^{-1}.}$$

Proof. Solve the hidden block equation

$$(zI - A_{QQ})x_Q = A_{QP}x_P + y_Q$$

for x_Q and substitute it into the retained equation. The resulting retained coefficient is $F_P(z)$. Invertibility of the full operator and of the hidden block implies invertibility of this Schur complement. Setting $y_Q = 0$ gives the stated identity as an operator on $\operatorname{Ran} P$. Equivalently, this is the block inverse formula. □

The self-energy

$$\Sigma_P(z) = A_{PQ}(zI - A_{QQ})^{-1}A_{QP}$$

contains the exact effect of the eliminated packets. A bare truncation A_{PP} discards this term and is generally not exact.

13.1 Time-domain memory

For the homogeneous system $\dot{x} = Ax$, variation of constants gives

$$x_Q(t) = e^{tA_{QQ}}x_Q(0) + \int_0^t e^{(t-r)A_{QQ}}A_{QP}x_P(r)\,dr.$$

Substitution yields

$$\dot{x}_P(t) = A_{PP}x_P(t) + A_{PQ}e^{tA_{QQ}}x_Q(0) + \int_0^t \mathcal{K}(t-r)x_P(r)\,dr,$$

where

$$\boxed{\mathcal{K}(t) = A_{PQ}e^{tA_{QQ}}A_{QP}.}$$

Theorem 13.2 (Memory/self-energy correspondence)**.** *For finite complex block matrices, assume* $\operatorname{Re} z > \operatorname{Re}\lambda$ *for every* $\lambda \in \sigma(A_{QQ})$*. Then the following improper integral is absolutely convergent and*

$$\boxed{\int_0^\infty e^{-zt}\mathcal{K}(t)\,dt = A_{PQ}(zI - A_{QQ})^{-1}A_{QP} = \Sigma_P(z).}$$

Proof. Put $B = A_{QQ}$ and $T_z(t) = e^{-zt}e^{tB} = e^{t(B-zI)}$. The generalized-eigenspace decomposition in finite dimensions writes T_z as a finite sum of constant matrix coefficients times $t^k e^{(\lambda-z)t}$. This follows by expanding the nilpotent part on each generalized eigenspace; equivalently, apply the decomposition to left multiplication by B on the matrix algebra and evaluate the exponential at I. The strict spectral inequality makes every such term absolutely integrable on $(0,\infty)$ and forces it to tend to zero. Hence T_z is Bochner integrable and $T_z(t) \to 0$, including when B is nonnormal.

Let $J = \int_0^\infty T_z(t)\,dt$. Integrating both derivative identities $T_z' = (B - zI)T_z = T_z(B - zI)$ and using $T_z(0) = I$ gives

$$(zI - B)J = J(zI - B) = I.$$

Thus $J = (zI - B)^{-1}$. Continuous left and right multiplication carry the Bochner integral through the two rectangular transfer blocks, yielding the claim. □

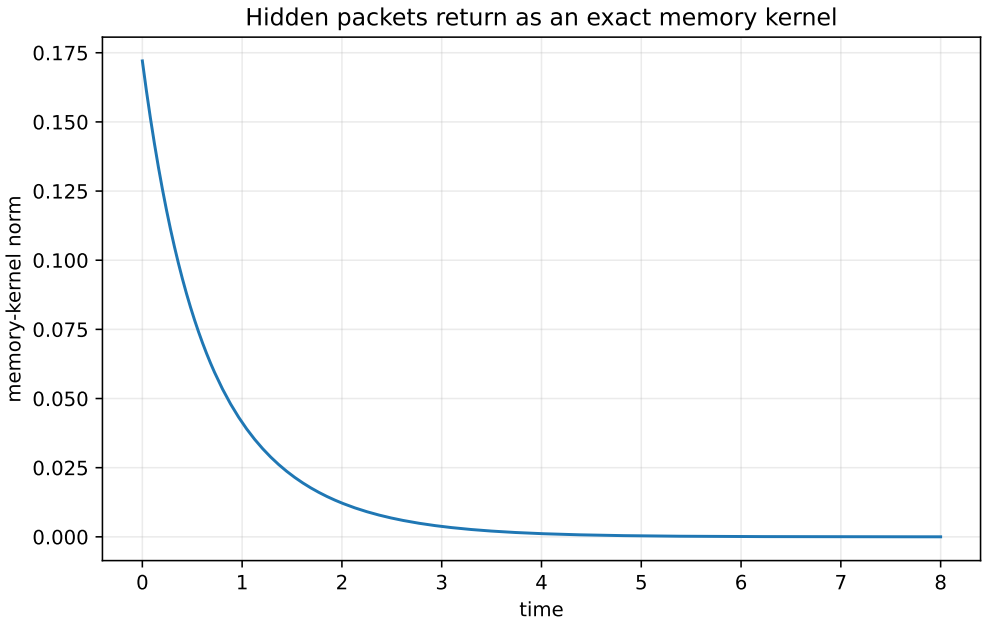


Figure 18: The norm of the exact memory kernel produced by eliminated packets.

14 Survival and hazard semigroups

Let A be a finite subgenerator: its off-diagonal entries are nonnegative and its column sums are nonpositive. These sign conditions alone allow nonabsorbing states; eventual absorption is a separate hypothesis. Let p_0 be an initial transient-state distribution. Define

$$p(t) = e^{tA}p_0, \qquad S(t) = \mathbf{1}^T p(t).$$

The exit-rate row vector is

$$a^T = -\mathbf{1}^T A \geq 0.$$

Theorem 14.1 (Semigroup survival identities)**.** *For $t \geq 0$, the finite-time exit density and, wherever $S(t) > 0$, the hazard are*

$$\boxed{f(t) = a^T e^{tA}p_0 = -S'(t), \qquad h(t) = \frac{f(t)}{S(t)}.}$$

Consequently, S is nonincreasing and

$$S(t) = \exp\left(-\int_0^t h(r)\,dr\right)$$

whenever $S(t) > 0$. The density may be defective: its total mass is $1 - \lim_{t\to\infty} S(t)$. It is a probability density when absorption is eventual, so that this limit is zero.

Proof. Differentiate:

$$S'(t) = \mathbf{1}^T A e^{tA}p_0 = -a^T e^{tA}p_0.$$

The hazard identity is the definition of conditional instantaneous exit rate. Integrating $(\log S)' = -h$ gives the final formula. □

15 One common operator language

The preceding constructions are not identical, but they use the same packet operations. The domain dictionary is summarized in table 1.

For clarity, the finite-dimensional holomorphic calculus used below has the following canonical definition. For A acting on $\mathbb{C}^d$, its generalized eigenspaces $E_\mu = \ker(A - \mu I)^d$ form a direct-sum decomposition over the eigenvalues. Define the unique linear operator $F(A)$ by

$$F(A)|_{E_\mu} = \sum_{k=0}^{d-1} \frac{F^{(k)}(\mu)}{k!}(A - \mu I)^k\Big|_{E_\mu}.$$

The formula is independent of a basis and of increasing the truncation index: the extra nilpotent powers vanish. It depends only on the finite Taylor jets of F at the spectrum. If F is holomorphic on an open neighborhood U of the spectrum, each eigenvalue admits a radius $r_\mu > 0$ with $\overline{D(\mu, r_\mu)} \subset U$, and Cauchy's formula gives

$$\frac{F^{(k)}(\mu)}{k!} = \frac{1}{2\pi i}\oint_{|z-\mu|=r_\mu} \frac{F(z)}{(z-\mu)^{k+1}}\,dz.$$

Consequently the same canonical operator is obtained from any such local Cauchy circles, even when U is disconnected. This is the standard finite Taylor-jet definition of the holomorphic matrix function.

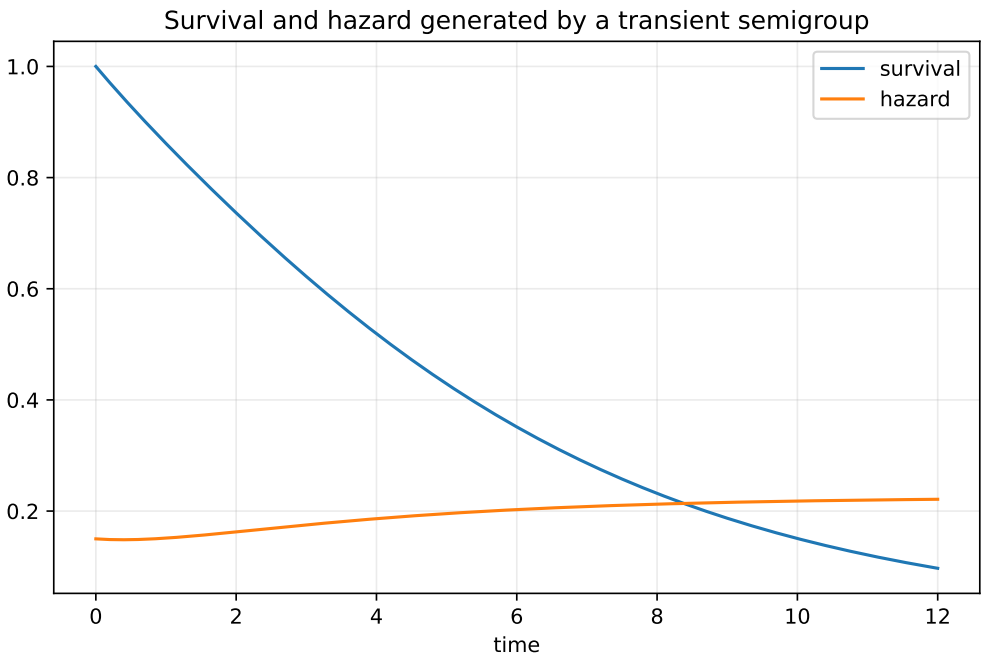


Figure 19: Survival and instantaneous hazard produced by one transient transfer semigroup.

Realization	Seed	Transfer	Packet/reconstruction law
Markov	point mass δ_i	K^n or e^{tQ}	state lumping and $CK^n = \bar{K}^n C$
Conditional	random variable	$P_{\mathcal{Q}}$	nested partitions and tower law
Bayesian	prior p	$\mathcal{B}_\lambda$	packet evidences and normalized synthesis
Fractional	spectral source	e^{-tL^α}	spectral projectors P_j
PDE	source f or δ_y	resolvent/Green operator	eigenmode or spatial packet synthesis
Feynman–Kac	terminal data f	$(\mathrm{diag}(e^{-V})K^T)^n$	weighted path reconstruction
Latent system	retained state	Feshbach/self-energy	hidden packets return as memory
Survival	initial transient law	e^{tA}	sum over transient packets

Table 1: The transfer-semigroup dictionary in the common operator language.

Theorem 15.1 (Naturality under exact intertwining)**.** *For finite complex matrices, let F be holomorphic on a neighborhood of the union of the spectra of A_f and A_c. Real matrices are treated by complexification. If*

$$CA_f = A_cC,$$

then

$$\boxed{CF(A_f) = F(A_c)C.}$$

In particular, exact generator refinement propagates simultaneously to semigroups, resolvents, integer powers, and analytic transfer functions.

Proof. The intertwining identity implies $C(A_f - \mu I)^k = (A_c - \mu I)^k C$ for every $k \geq 0$. It therefore sends the generalized μ-eigenspace of A_f into that of A_c. Use a common truncation index at least as large as both dimensions in the canonical formula above. Multiplying its finite sum by C now gives the desired identity on every generalized eigenspace. The direct-sum decomposition gives it on the full space. The argument covers nilpotent Jordan parts and uses only local spectral Taylor jets; it does not require one Taylor series to converge on both spectra. The zero-dimensional case is immediate. □

Remark 15.2 (Where nonlinearity enters)*.* Bayesian conditioning does not fit the preceding linear functional calculus because normalization depends on the current state. It nevertheless obeys an exact composition law and has a closed packet reconstruction. Paper III supplies the general nonlinear coordinate theorem; the Bayesian map is a particularly transparent probabilistic realization.

16 Numerical certificates

All calculations in this release were run on CPU. No GPU timing or performance claim is made. Floating-point residuals are implementation checks and are not substitutes for the symbolic proofs.

Numerical Certificate 16.1 (Test suite)*.* The complete release passes

$$\boxed{54/54 \text{ automated tests.}}$$

The tests cover projector reconstruction, packet-block semigroup composition, exact and approximate Markov lumping, Bayesian composition and flow, the exact Bayesian refinement criterion, information-loss factorization, likelihood-profile partitions, conditional-expectation towers, Hermite orthogonality, fractional semigroups, square-root subordination, fractional intertwining and semigroup-defect bounds, discrete Green kernels, PDE convergence, Feynman–Kac enumeration, Schur–Feshbach elimination, memory kernels, refinement defects, hidden Markov filtering, and survival curves.

17 Remaining conjectural extensions

The finite theory closes several operator identities but does not settle the corresponding universal continuum questions.

Conjecture 17.1 (Transfer-semigroup realization)**.** Every sufficiently regular realization of the seed calculus whose transfer stage is local in time admits a closed generator on a completed state space, and the packet analysis maps form a compatible system for its semigroup or evolution family.

Conjecture 17.2 (Uniform refinement control)**.** For a nested family of physically or analytically meaningful discretizations, there exist norms and constants independent of resolution such that

$$\left\| C_\ell e^{tA_{\ell+1}} - e^{tA_\ell} C_\ell \right\| \leq M\, t\, \|C_\ell A_{\ell+1} - A_\ell C_\ell\|$$

Certificate	Value
Packet-block semigroup composition residual	2.554×10^{-15}
Operator reconstruction residual	0
Random-walk mass error	0
Exact Markov generator-lumping defect	5.551×10^{-17}
Exact finite-horizon lumping error	2.220×10^{-16}
Perturbed ten-step lumping error	6.523×10^{-3}
Linear defect bound for that example	2.000×10^{-1}
Bayesian composition residual	5.551×10^{-17}
Bayesian packet reconstruction residual	0
Bayesian refinement commutation residual	1.110×10^{-16}
Bayesian defect-factorization residual	7.286×10^{-17}
Bayesian flow generator residual	1.959×10^{-11}
Conditional-expectation tower residual	0
Hermite Gram residual	9.992×10^{-15}
Fractional semigroup residual	2.082×10^{-17}
Maximum square-root subordination error	1.665×10^{-16}
Exact fractional-naturality residual	0
Fractional-power defect / gap certificate	0.51059/0.95079
Fractional-semigroup defect / certificate	0.12969/0.44480
Observed low-mode PDE convergence order	1.999854
Feynman–Kac enumeration residual	1.665×10^{-16}
Feshbach projected-resolvent residual	5.551×10^{-17}
Memory Laplace-transform residual	6.939×10^{-18}
Duhamel refinement identity residual	2.108×10^{-16}
Survival monotonicity violation	0
Hidden-Markov normalization error	1.110×10^{-16}

Table 2: Selected results from the machine-readable numerical record.

on every bounded time interval. Summable generator defects would then produce a coherent inverse-limit evolution.

Conjecture 17.3 (Latent-packet closure)**.** For a stable infinite packet system, exact elimination of unresolved packets produces a causal memory kernel whose Laplace transform is the limiting packet self-energy. A Markovian closure exists precisely when this self-energy is constant, or can be represented by finitely many additional state packets.

Boundary 17.4 (What remains outside Paper VI)*.* The following issues are not proved here:

1. existence and uniqueness for arbitrary stochastic differential equations;
2. posterior consistency or asymptotic normality in infinite statistical models;
3. domain invariance for unbounded fractional operators;
4. tightness and compactness required for stochastic continuum limits;
5. nonlinear PDE well-posedness beyond the stated finite and linear models;
6. an optimal packet choice for a general Markov, Bayesian, or PDE problem.

These belong to later realization-specific work or to external analytic theory.

18 Computational implementation

The accompanying Python package implements the formulas and finite models used in the paper.

`projector_transfer_semigroups`

The computational organization mirrors the paper:

`packets.py`	projectors, blocks, reconstruction, block composition
`semigroup.py`	matrix semigroups, defects, Feshbach operators, memory
`markov.py`	random walks, exact lumping, survival
`bayes.py`	normalized updates, packet evidence, conjugacy, flows, HMM
`conditional.py`	conditional expectation and Gaussian chaos
`fractional.py`	spectral powers, subordination, and fractional refinement certificates
`pde.py`	Dirichlet Laplacians, Green kernels, eigenvalue error
`feynman_kac.py`	weighted transfer and exhaustive path enumeration.

18.1 Representative algorithms

Listing 1: Normalized finite Bayesian update.

```
def bayes_update(prior, likelihood):
    p = normalize(prior)
    evidence = likelihood @ p
    if evidence <= 0:
        raise ValueError("positive evidence required")
    return likelihood * p / evidence
```

Listing 2: Packet block composition.

```
def block_composition(left, right):
    result = []
    for beta in range(len(left)):
        row = []
        for alpha in range(len(left)):
            value = sum(
                left[beta][gamma] @ right[gamma][alpha]
                for gamma in range(len(left))
            )
            row.append(value)
        result.append(row)
    return result
```

Listing 3: Finite Schur-Feshbach effective operator.

```
def feshbach_effective_operator(A, retained, z):
    App, Apq, Aqp, Aqq = split_blocks(A, retained)
    return z*I - App - Apq @ inv(z*I - Aqq) @ Aqp
```

The full implementation, tests, tables, figure-generation code, and exact numerical record are included with the release rather than embedded in the typeset paper.

19 Conclusion

The transfer-semigroup realization gives one common operator language to several subjects that are often presented separately. Markov propagation is linear transfer of a probability seed. Conditional expectation is stochastic projection. Bayesian conditioning is normalized nonlinear transfer, and its exact finite coarsening law is now characterized by packetwise likelihood sufficiency. Fractional calculus is spectral packet reweighting, and bounded positive refinement defects now transfer to fractional powers and fractional semigroups by explicit resolvent and Duhamel estimates. A Green operator reconstructs a field from source packets. Feynman–Kac propagation adds path weights to stochastic transfer. Schur–Feshbach elimination shows that hidden packets are not destroyed; they return through self-energy and memory. Survival and hazard are scalar reconstructions of a transient semigroup.

The central structural identity is

$$\boxed{\text{generator defect} \longrightarrow \text{Duhamel semigroup defect} \longrightarrow \text{controlled refinement error.}}$$

Together with exact packet reconstruction, this makes the semigroup paper the natural bridge from the foundational calculus to the quantum, computational, and arithmetic realizations that follow.

A Lean verification and scope

Fifth checked block, 6 September 2026. The release contains 341 explicit Lean theorem declarations across 47 proof modules, including 31 additional declarations since the fourth checked block. The full build and namespace audit pass with Lean 4.34.0-rc2 and pinned mathlib. The audit inspects 782 declarations, including definitions and generated auxiliaries; only `propext`, `Classical.choice` and `Quot.sound` are allowed. These are different counting units from the paper's 33 theorem-like results.

The ledger has **33 complete, zero partial and zero pending results**. The last two results, 13.2 and 15.1, are now complete. The three separately stated conjectures remain open and are excluded from this count.

"Complete" means the stated scope follows from the named checked targets and the ordinary specializations recorded in the ledger. Prose-to-Lean correspondence is manually reviewed. The fractional targets use complex Hilbert spaces; complexification for real operators is an ordinary mathematical specialization, not a separately checked Lean construction. Completion of the theorem ledger does not certify every prose sentence, numerical figure or open conjecture.

Result	Manuscript statement	Lean scope
1.2	Finite semigroup law	Complete
2.1	Packet semigroup convolution	Complete
2.2	Reducing packets	Complete
3.1	Exact Duhamel refinement identity	Complete
3.2	Norm bound	Complete
3.3	Resolvent refinement identity	Complete
4.1	Finite Markov semigroup	Complete
5.1	Exact lumpability	Complete
5.2	Finite-horizon defect formula	Complete
6.1	Finite conditional-expectation projector	Complete
6.2	Tower law and martingale packets	Complete
7.1	Independent-likelihood composition	Complete
7.2	Bayesian packet reconstruction	Complete
7.3	Likelihood perturbation bound	Complete
8.1	Exact effective-likelihood factorization	Complete
8.2	Bayesian refinement criterion	Complete
8.4	Bayesian information-loss packet	Complete
8.5	Nested Bayesian packet tower	Complete
8.6	Canonical coarsest likelihood-resolving partition	Complete
9.1	Bayesian nonlinear semigroup	Complete
9.3	Prediction-update reconstruction	Complete
10.1	Fractional packet calculus	Complete
10.2	Poisson subordination	Complete
10.3	Fractional intertwining theorem	Complete
10.4	Fractional semigroup refinement	Complete
10.5	Fractional refinement towers	Complete
11.1	Exact discrete Green reconstruction at grid nodes	Complete
11.2	Second-order low-mode convergence	Complete
12.1	Discrete Feynman–Kac path formula	Complete
13.1	Schur–Feshbach projected resolvent	Complete
13.2	Memory/self-energy correspondence	Complete
14.1	Semigroup survival identities	Complete
15.1	Naturality under exact intertwining	Complete

General nonnormal memory (13.2). The actual damped exponential is decomposed into finitely many terms of the form $t^k e^{(\mu-z)t}$. Integrability and convergence to zero follow from $\operatorname{Re}\mu < \operatorname{Re} z$. The formal proof uses generalized eigenspaces of left multiplication on the matrix algebra, so nilpotent Jordan parts are included. Both inverse identities follow by integrating the two exponential derivative formulas. A continuous linear sandwich map gives the literal rectangular matrix memory integral and the actual inverse. No normality or externally supplied growth bound is required.

Canonical holomorphic naturality (15.1). The standard finite Taylor-jet calculus defined before the theorem is constructed using the direct sum of generalized eigenspaces. Uniqueness, cutoff stability and exact rectangular naturality are proved. Cauchy's formula identifies the local coefficients with actual circle integrals; admissible radii exist in every stated open holomorphic neighborhood, and changing them leaves the canonical operator unchanged. The argument permits disconnected neighborhoods and different matrix dimensions. It makes no assertion about general Banach-algebra contour deformation.

Fractional results (10.3–10.5). The inherited fourth block proves the scalar beta normalization, the Lebesgue integral representation of actual positive fractional powers, the literal resolvent-defect formula, both printed norm constants and exact intertwining. The rectangular complex-Hilbert-space semigroup proof substitutes those constants into Duhamel's identity. The varying-space tower proof gives summability of fractional-defect norms and uniform convergence of time-defect norm series on bounded time intervals. Here "summable" means summable in norm. Operators with unrelated codomains are not added together; a separate fixed-space theorem proves operator-series convergence when such a sum is defined.

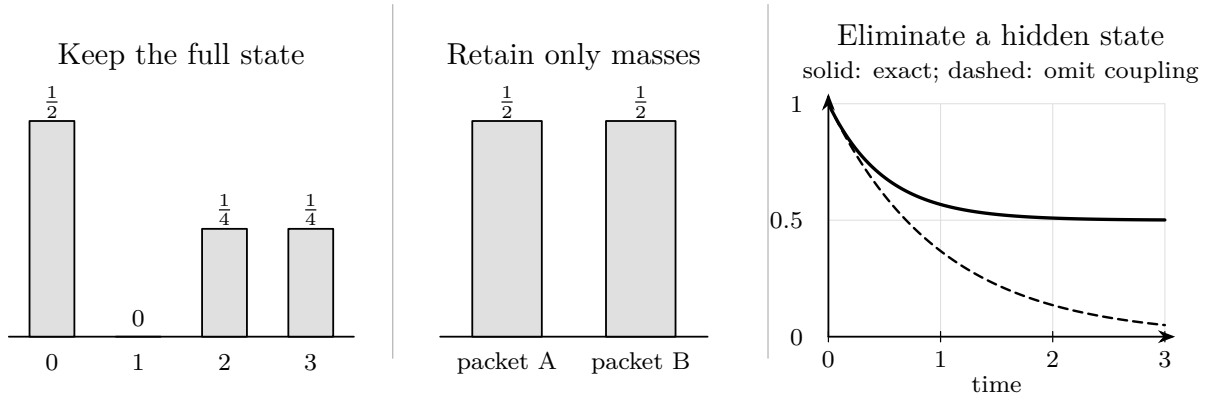
Verification and numerical evidence. The full pinned build, project-wide axiom audit, all 54 companion tests and six verifier controls pass. The 54 tests comprise the original 37 and 17 added boundary and validation cases. The 19 numerical figure pairs are unchanged. The earlier 79-field numerical comparison remains historical evidence, not a rerun in this block. Machine-readable source contracts, exact theorem targets, full logs and previous checked ledgers accompany the source archive. Numerical tests are finite regressions and are not substitutes for the Lean proofs.

PAPER VI / A WORKED VISUAL ATLAS

Three ways to reduce a model

Main text: Sections 1–3, 8, and 13. Read the three operations separately.

This appendix follows small objects that can be computed completely. A four-state chain connects packet transfer, Markov aggregation, and Bayesian conditioning. A two-state coupled evolution exposes memory. A three-node Dirichlet problem makes the Green-operator normalization visible. These models are related by the paper's language, not by pretending that their state spaces are identical.



Left and center use one probability vector and its two block masses. The right panel is a separate retained/hidden model, developed on V – 22. Every curve is drawn from an explicit formula.

Coordinates retain; aggregation forgets; elimination remembers

Keeping every component $P_a x$ is an exact change of representation when $\sum_a P_a = I$. Reconstruct by summing the components. In contrast, replacing (p_0, p_1, p_2, p_3) by $(p_0 + p_1, p_2 + p_3)$ forgets how mass was distributed inside either pair. That information can matter for later observations.

Eliminating a hidden state is different again. Its coordinates disappear from the displayed state, but the exact retained equation acquires a history integral. Simply deleting the coupling produces the dashed curve above, not the original evolution.

Question	Worked route	Pages
When is coarse evolution exact?	Aggregate columns, then test intertwining.	V – 4–V – 6
When is a coarse posterior exact?	Compare two hidden shapes with the same masses.	V – 9–V – 13
What changes under a fractional power?	Keep modes, change spectral rates.	V – 15–V – 17
What returns after elimination?	Compute the self-energy and memory kernel.	V – 21–V – 25

Read this as: The paper supplies exact identities and conditional bounds. Whether a particular coarse model satisfies their hypotheses is a separate calculation.

READ BEFORE MULTIPLYING / CONVENTIONS

A dictionary with orientations

Main text: Sections 2, 4, 6, 7, and 12. Column probabilities and backward observables.

An apparent disagreement between two transfer formulas often comes from acting on different objects. Throughout this appendix, probabilities are columns and observables are functions written as columns. Their pairing is $f^{\mathsf{T}}p$. The transpose distinguishes the forward action on distributions from the backward action on observations.

Object	Exact meaning	What to check
p	A nonnegative column with $\mathbf{1}^{\mathsf{T}}p = 1$.	It is not a vector of function values.
Q	Continuous Markov generator.	Off-diagonal entries are nonnegative and columns sum to zero.
K	One discrete stochastic step.	$\mathbf{1}^{\mathsf{T}}K = \mathbf{1}^{\mathsf{T}}$ and $K_{ij} \geq 0$.
$K^{\mathsf{T}}f$	Expected next observable given the current state.	The same dynamics, with the opposite action on the dual object.
P_a	A coordinate packet projector.	All packets together reconstruct the full state.
C	Sums fine probability masses into coarse blocks.	Usually rectangular and not invertible.
E_π	Conditional expectation of observables.	A weighted average, self-adjoint for the π inner product.
$\mathcal{B}_\lambda$	$\mathrm{diag}(\lambda)p/(\lambda^{\mathsf{T}}p)$.	Nonnegative likelihood and positive total evidence.
L	A nonnegative self-adjoint operator.	Its evolution is e^{-tL}, not e^{tL}.
$R(z)$	$(zI - A)^{-1}$.	z belongs to the resolvent set.
$\mathcal{K}(t)$	The memory kernel after hidden-state elimination.	It is not the transition matrix K.

One convention test

For

$$K_2 = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \qquad p = e_0, \qquad f = (2,1)^{\mathsf{T}},$$

we obtain $K_2 p = (3/4, 1/4)^{\mathsf{T}}$ and

$$f^{\mathsf{T}}K_2 p = 7/4 = (K_2^{\mathsf{T}}f)_0.$$

With departure weights $w_i = e^{-V(i)}$, the backward Feynman–Kac operator is $\mathrm{diag}(w)K_2^{\mathsf{T}}$. Its forward dual is $K_2\,\mathrm{diag}(w)$. Interchanging these factors changes when the weight is applied.

> **Read this as:** Always name the object being transported. The transpose and the position of the likelihood or potential are mathematical choices, not typographic details.

CONTINUOUS TRANSFER / THE ENTIRE SOLUTION

A clock generated by one matrix

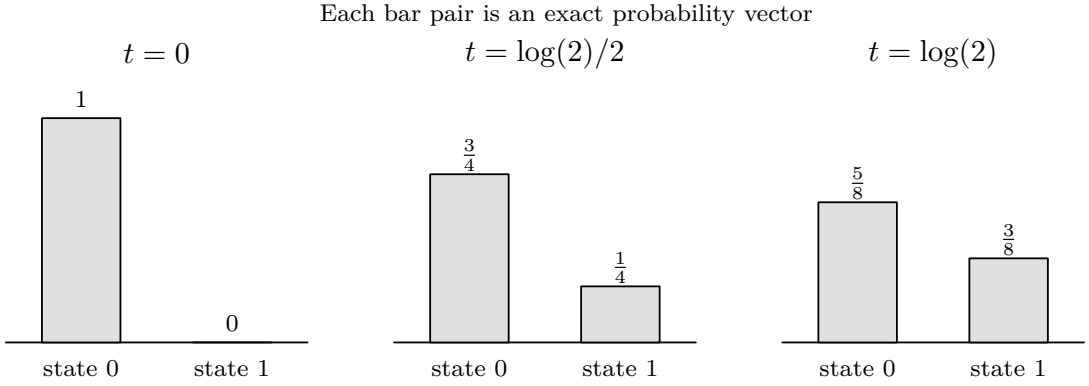
Main text: Theorems 1.2 and 4.1. Finite matrix exponentials and Markov positivity.

Take the generator

$$G = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The total-mass direction $(1, 1)$ has eigenvalue zero. The contrast $(1, -1)$ has eigenvalue -2. Resolving those two directions gives the full transfer, not only its first-order approximation:

$$T(t) = e^{tG} = \frac{1}{2}\begin{pmatrix} 1 + e^{-2t} & 1 - e^{-2t} \\ 1 - e^{-2t} & 1 + e^{-2t} \end{pmatrix}.$$



A point seed at state zero approaches the uniform distribution. The displayed times were selected so that every bar label is rational. This is continuous-time evolution, not a simulated trajectory.

Two checks explain the mechanism

The constant mode is unchanged, while each elapsed interval multiplies the contrast by e^{-2t}. Consequently two intervals multiply their contrast factors:

$$e^{-2s}e^{-2t} = e^{-2(s+t)}, \qquad T(s)T(t) = T(s+t).$$

For $t \geq 0$, the entries are nonnegative and every column sums to one. Differentiation gives $T'(t) = GT(t)$. These three facts separately check the initial condition, the evolution equation, and preservation of probability.

At $t = \log 2$, the retained first coordinate is $5/8$. It would instead be $1/2$ under the isolated scalar equation $\dot{x} = -x$. The discrepancy is the mass that can leave state zero and return. On V – 22, we remove the second coordinate correctly and recover exactly the same $5/8$ using a memory integral.

> **Read this as:** The semigroup lives on the complete state. One diagonal entry of a semigroup need not itself satisfy a scalar semigroup law.

FOUR-STATE MODEL / COORDINATES AND MODES

One evolution, two packet choices

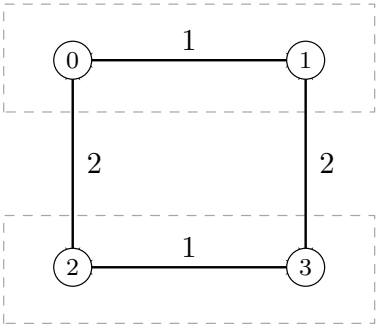
Main text: Theorem 2.1, Corollary 2.2, and the packet path expansion.

Our main finite model has generator

$$Q = \begin{pmatrix} -3 & 1 & 2 & 0 \\ 1 & -3 & 0 & 2 \\ 2 & 0 & -3 & 1 \\ 0 & 2 & 1 & -3 \end{pmatrix}.$$

The graph shows rates between actual states. The four normalized sign vectors on the right form an orthonormal basis u_j. Their projectors $P_j = u_j u_j^{\mathsf{T}}$ sum to the identity and satisfy $P_i P_j = 0$ for $i \neq j$.

Off-diagonal rates of Q



Top pair A; bottom pair B

Mode signs (each divided by 2)

	0	1	2	3	rate
constant	1	1	1	1	0
within	1	−1	1	−1	−2
between	1	1	−1	−1	−4
mixed	1	−1	−1	1	−6

State coordinates have off-diagonal transfers. Spectral packets reduce the generator and evolve independently. The two dashed state groups are the coarse partition used next.

The same evolution has a closed formula

Writing $q = e^{-2t}$ gives

$$e^{tQ} = P_0 + qP_1 + q^2 P_2 + q^3 P_3.$$

At $q = 1/2$, the point seed e_0 becomes $(15, 5, 9, 3)^{\mathsf{T}}/32$. The polynomial identity $T(q)T(r) = T(qr)$ checks composition exactly, without numerical matrix exponentiation.

Any other complete packet family still has exact block convolution. For the coordinate groups A and B,

$$P_b e^{(s+t)Q} P_a = \sum_{c \in \{\mathrm{A},\mathrm{B}\}} (P_b e^{sQ} P_c)(P_c e^{tQ} P_a).$$

The intermediate packet cannot generally be omitted. Only a reducing family eliminates all cross terms.

For later discrete calculations define $K = I + Q/8$. This matrix is stochastic, but *it is not* $e^{Q/8}$. It is the step kernel in the uniformization identity $e^{tQ} = e^{-8t} \sum_{n \geq 0} (8t)^n K^n / n!$.

> **Read this as:** Changing packet coordinates does not change the evolution. Replacing continuous time by one discrete step does.

EXACT LUMPABILITY / AGGREGATE COLUMNS

When two states are truly equivalent

Main text: Theorem 5.1. The hypothesis is an intertwining identity.

Group states $0, 1$ into A and states $2, 3$ into B. The aggregation map and the candidate coarse kernel are

$$C = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad \bar{K} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

A fine state in A has probability $1/4$ of entering B in one step, regardless of which fine state in A was occupied. The analogous statement holds for B.

$8K$

	0	1	2	3
0	5	1	2	0
1	1	5	0	2
2	2	0	5	1
3	0	2	1	5

CK: aggregate transition columns

	0	1	2	3
A	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
B	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$

$$\bar{K} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}$$

Fine trajectories differ. Coarse predictions agree.

Columns 0 and 1 agree; columns 2 and 3 agree.

The fine columns are visibly different. Their sums over each target packet agree within each source packet. That equality is the reason a two-state description works.

A one-step calculation controls every discrete horizon

The displayed columns give $CK = \bar{K}C$. Multiplication then yields

$$CK^n = \bar{K}^n C \qquad (n \geq 0).$$

For example, e_0 and e_1 have the same coarse initial vector $(1, 0)^\mathsf{T}$. Their fine evolutions differ, but their coarse vectors agree at every step. No choice of the within-A conditional distribution is needed for the prediction.

The continuous generator also closes:

$$CQ = \bar{Q}C, \qquad \bar{Q} = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}.$$

The Duhamel identity therefore gives $Ce^{tQ} = e^{t\bar{Q}}C$. This is a second consequence of the corresponding generator identity, not an inference that K is a fixed-time sample of Q.

Aggregation is still not invertible. The theorem says the discarded details are irrelevant for *these predictions*. It does not say they are irrelevant to every later likelihood. The observation counterexample on V – 10 uses this exact same partition.

> **Read this as:** A partition is not automatically a valid Markov model. The aggregate transition columns must be constant within each source block.

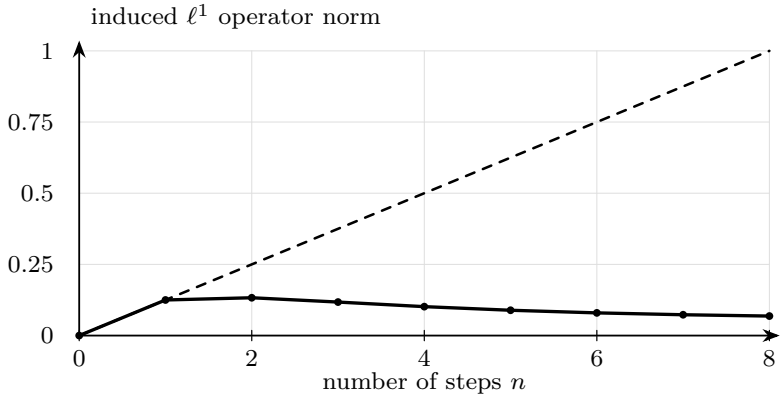
DEFECTS / A GUARANTEE WITHOUT A FIT

A broken lump and its error budget

Main text: Theorems 3.1, 3.3, and 5.2. Continuous and discrete defect identities.

Modify only column zero of K: move $1/16$ of its mass from target state zero to target state two. The perturbed matrix K_ε remains stochastic, but its aggregate column now differs from column one. Relative to the unchanged coarse kernel,

$$\Delta = CK_\varepsilon - \bar{K}C = \begin{pmatrix} -1/16 & 0 & 0 & 0 \\ 1/16 & 0 & 0 & 0 \end{pmatrix}, \qquad \|\Delta\|_1 = 1/8.$$



$\|\Delta\|_1 = 1/8$

solid: exact defect
dashed: $n/8$

The bound is a guarantee, not a fitted trend.

The solid values are computed with rational matrix powers. The dashed certificate uses only stochasticity and the exact one-step defect. It need not be sharp.

Telescoping tells us where every error originates

For $n \geq 0$,

$$CK_\varepsilon^n - \bar{K}^n C = \sum_{j=0}^{n-1} \bar{K}^{n-1-j} \Delta K_\varepsilon^j.$$

Each summand propagates one defect between an earlier fine evolution and a later coarse evolution. Column-stochastic matrices have induced ℓ^1 norm one, so the norm is at most $n/8$. The empty sum at $n = 0$ is zero.

The continuous counterpart is equally explicit. With fine generator -1, coarse generator -2, and $C = 1$, the generator defect is 1 and

$$e^{-t} - e^{-2t} = \int_0^t e^{-2(t-r)} e^{-r}\, dr.$$

Its resolvent version is $1/(z+1) - 1/(z+2) = 1/((z+1)(z+2))$ for $z > 0$.

> **Read this as:** A small defect does not disappear. The identity specifies how it accumulates, and the norm assumptions turn that identity into a finite-horizon certificate.

OBSERVABLES / WEIGHTED ORTHOGONALITY

Averaging is not summing

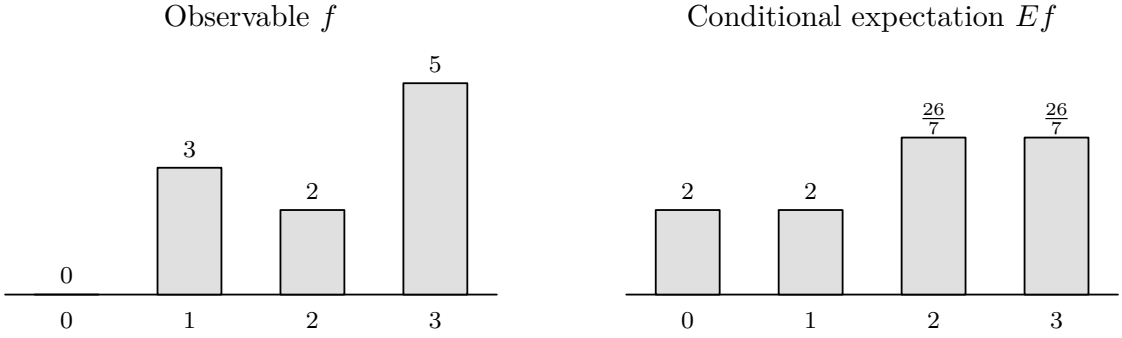
Main text: Theorem 6.1. Strictly positive weights are part of the setting.

For this observable example, choose the fixed probability weights $\pi = (1, 2, 3, 4)/10$. They are not being identified with the stationary law of the earlier symmetric chain. On the same two-block partition, conditional expectation is

$$E = \begin{pmatrix} 1/3 & 2/3 & 0 & 0 \\ 1/3 & 2/3 & 0 & 0 \\ 0 & 0 & 3/7 & 4/7 \\ 0 & 0 & 3/7 & 4/7 \end{pmatrix}.$$

It replaces function values by their *weighted block average*. In particular, $f = (0, 3, 2, 5)^{\mathsf{T}}$ becomes $Ef = (2, 2, 26/7, 26/7)^{\mathsf{T}}$.

Weights $\pi = (1, 2, 3, 4)/10$; the block values are 2 and 26/7.



The bars are function values, not state probabilities. Their total need not be one. The block masses 3/10 and 7/10 determine the denominators in the averaging formula.

Which inner product makes this an orthogonal projection?

Put $W = \mathrm{diag}(\pi)$. Direct multiplication checks

$$E^2 = E, \qquad WE = E^{\mathsf{T}}W, \qquad E\mathbf{1} = \mathbf{1}.$$

The middle equation is weighted self-adjointness. Ordinary symmetry fails: $E_{01} = 2/3$ whereas $E_{10} = 1/3$. Testing $E = E^{\mathsf{T}}$ would therefore reject a correct conditional-expectation projector.

The range consists exactly of functions constant on each block. The error $f - Ef$ has weighted mean zero inside every block, so it is weighted-orthogonal to every block-constant function. This is the finite projection interpretation of conditional expectation.

In contrast, C acts on probability columns by summing masses. The shapes of C and E differ, and their domains have different interpretations. The notation "coarsening" alone does not specify which operation is intended.

> **Read this as:** An exact weighted projector can be oblique in the unweighted Euclidean picture. Keep the measure and the inner product in the model.

MARTINGALE PACKETS / THREE EXACT PIECES

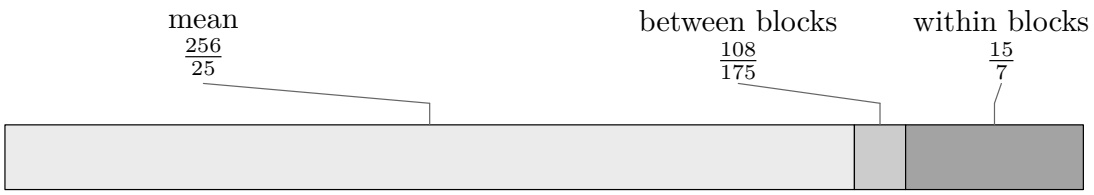
Where the variance goes

Main text: Theorem 6.2. Nested weighted projections and their differences.

Keep the weights and observable from V–7. Let $E_0 f$ be the constant global mean, let $E_1 = E$ average within A and B, and let $E_2 = I$ retain every value. Define

$$D_0 = E_0, \qquad D_1 = E_1 - E_0, \qquad D_2 = I - E_1.$$

Then $D_0 f$ is the mean, $D_1 f$ describes the contrast between block means, and $D_2 f$ is the fluctuation left inside each block.



$$D_0 f = \begin{bmatrix} \frac{16}{5} \\ \frac{16}{5} \\ \frac{16}{5} \\ \frac{16}{5} \end{bmatrix} \qquad D_1 f = \begin{bmatrix} -\frac{6}{5} \\ -\frac{6}{5} \\ \frac{18}{35} \\ \frac{18}{35} \end{bmatrix} \qquad D_2 f = \begin{bmatrix} -2 \\ 1 \\ -\frac{12}{7} \\ \frac{9}{7} \end{bmatrix}$$

The strip lengths are proportional to the three exact weighted energies. The between-block component is small but not zero. The vectors below specify the corresponding directions, not only their sizes.

The energy identity follows from actual orthogonality

The tower identities give $E_0 E_1 = E_1 E_0 = E_0$. Together with weighted self-adjointness, these imply

$$D_i D_j = 0 \ (i \neq j), \qquad \sum_{j=0}^{2} D_j = I.$$

For the chosen f, the total weighted energy is $\|f\|_\pi^2 = 13$ and the mean is $16/5$. Consequently

$$\mathrm{Var}_\pi(f) = 13 - \left(\frac{16}{5}\right)^2 = \frac{69}{25} = \frac{108}{175} + \frac{15}{7}.$$

The first term is the energy of the between-block contrast, and the second is the weighted within-block variance. This is a decomposition of one observable in one fixed weighted space.

Changing the probability weights between levels would change the inner products and the averaging operators. The displayed orthogonal decomposition would no longer follow merely from nesting the sets. Likewise, choosing unrelated projections can preserve the telescope $\sum D_j = I$ while losing orthogonality and the energy sum.

> **Read this as:** The differences between nested conditional expectations are not just bookkeeping. They are orthogonal, separately measurable parts of the original signal.

BAYESIAN TRANSFER / AN EXACT POSTERIOR

Evidence changes shape and mass

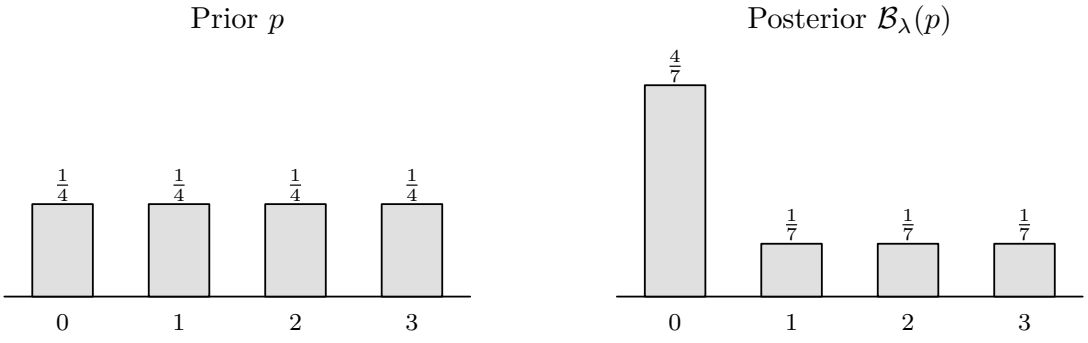
Main text: Theorems 7.1–7.3. Normalization, packet reconstruction, and positive evidence.

Start with $p = (1,1,1,1)^{\mathsf{T}}/4$ and likelihood weights $\lambda = (4,1,1,1)^{\mathsf{T}}$. The unnormalized vector is $(1,1/4,1/4,1/4)^{\mathsf{T}}$, its evidence is $Z = 7/4$, and

$$\mathcal{B}_\lambda(p) = \frac{(4,1,1,1)^{\mathsf{T}}}{7}.$$

Likelihood weights may be rescaled, so an evidence larger than one is not a probability claim. Multiplying all likelihoods by one common positive factor leaves this posterior unchanged.

$\lambda = (4,1,1,1)$; evidence $Z = 7/4$



The A packet becomes heavier and its internal shape changes. The B packet retains an equal internal split but loses total mass. These are different effects of the same likelihood.

Reconstruct from packet mass times packet shape

The unnormalized packet evidences are $Z_{\mathrm{A}} = 5/4$ and $Z_{\mathrm{B}} = 1/2$. The resulting masses and conditional shapes are

$$m_{\mathrm{A}} = 5/7, \quad q_{\mathrm{A}} = (4/5, 1/5, 0, 0)^{\mathsf{T}}, \qquad m_{\mathrm{B}} = 2/7, \quad q_{\mathrm{B}} = (0, 0, 1/2, 1/2)^{\mathsf{T}}.$$

Thus $m_{\mathrm{A}} q_{\mathrm{A}} + m_{\mathrm{B}} q_{\mathrm{B}}$ is exactly the displayed posterior. Keeping only the masses would not reconstruct the fine state.

There is no need to divide by a zero packet evidence. For likelihood $(0,0,1,1)$, packet A has mass zero and contributes the zero vector. Only the occupied B packet has a conditional probability vector. If the *total* evidence is zero, however, the Bayesian update is not defined and the implementation rejects the input.

> **Read this as:** Bayesian reconstruction uses two kinds of data: how much posterior mass a packet carries and how that mass is distributed inside the packet.

SUFFICIENCY / A COMPUTABLE FAILURE WITNESS

Two priors that fool the coarse model

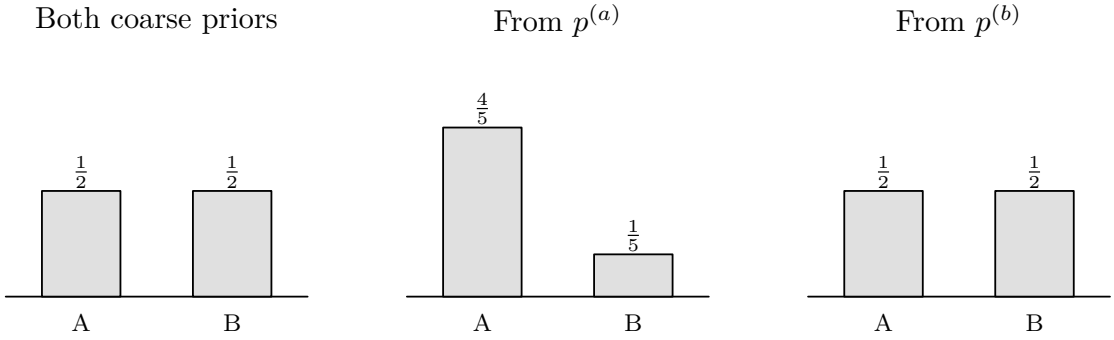
Main text: Theorem 8.2 and the one-packet exception.

Keep the same likelihood $\lambda = (4, 1, 1, 1)^{\mathsf{T}}$ and partition A= $\{0, 1\}$, B= $\{2, 3\}$. Compare

$$p^{(a)} = (1/2, 0, 1/2, 0)^{\mathsf{T}}, \qquad p^{(b)} = (0, 1/2, 1/2, 0)^{\mathsf{T}}.$$

Both have coarse prior $(1/2, 1/2)^{\mathsf{T}}$. A procedure given only that two-entry vector cannot distinguish them.

Hidden redistribution changes the packet-A posterior from 1/2 to 4/5.



After the identical fine likelihood is applied, the coarse posteriors are $(4/5, 1/5)$ and $(1/2, 1/2)$. A single deterministic update of the common coarse prior cannot produce both.

Why this witnesses the necessity of blockwise constancy

The A evidence for $p^{(a)}$ is 2, whereas it is 1/2 for $p^{(b)}$. Their B evidence is 1/2 in both cases. The posterior odds therefore distinguish the hidden distribution inside A.

For strictly positive fine and coarse likelihoods, and at least two coarse blocks, the manuscript proves the all-prior criterion

$$C\mathcal{B}_\lambda(p) = \mathcal{B}_{\bar{\lambda}}(Cp) \quad \text{for every } p \quad \Longleftrightarrow \quad \lambda_i = c\bar{\lambda}_a \quad (i \in I_a)$$

for one global $c > 0$. The global scale is invisible after normalization. A different arbitrary scale for each packet is not invisible: it changes posterior odds.

If all states are merged into a single packet, every normalized posterior coarsens to (1). That degenerate commutation holds for every likelihood and cannot identify any within-packet information. It is explicitly excluded from the nontrivial criterion.

> **Read this as:** A prior-dependent effective likelihood can always summarize the update. A fixed likelihood that works for every prior is a much stronger requirement.

BAYESIAN DEFECT / MORE THAN AN ERROR BAR

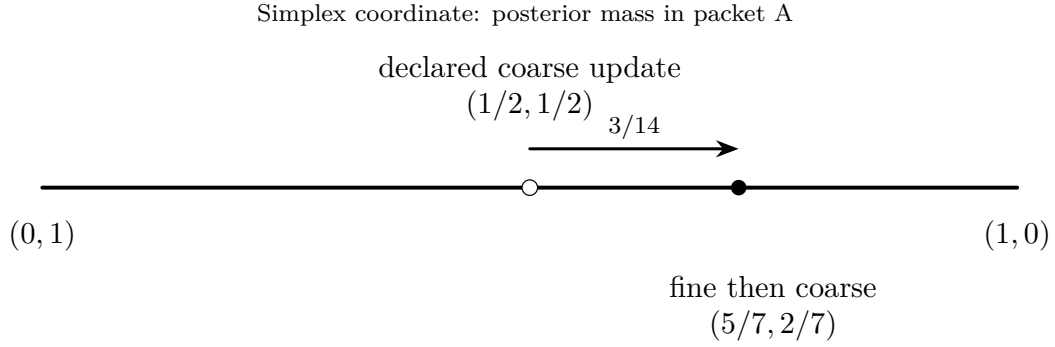
The exact information-loss vector

Main text: Theorems 8.1 and 8.4, and Corollary 8.6.

For a prior p, let $P = Cp$ and let $r_a(p)$ be the conditional average of the fine likelihood inside packet a. For the uniform prior and $\lambda = (4, 1, 1, 1)$,

$$P = (1/2, 1/2)^{\mathsf{T}}, \qquad r = (5/2, 1)^{\mathsf{T}}.$$

The exact coarse posterior is $\mathcal{B}_r(P) = (5/7, 2/7)^{\mathsf{T}}$. Suppose instead that the declared coarse likelihood is $\bar{\lambda} = (1, 1)$.



The posterior error lies along the mass-preserving direction of the two-state simplex. Its entries are opposite because both endpoints have total mass one.

Compute the factorization, including the null direction

Here $\bar{b} = \mathcal{B}_{\bar{\lambda}}(P) = P$, $\delta = r - \bar{\lambda} = (3/2, 0)^{\mathsf{T}}$, and $Z_r = 7/4$. Hence

$$\frac{1}{Z_r}\left(\operatorname{diag}(P) - \bar{b}P^{\mathsf{T}}\right)\delta = \frac{4}{7}\begin{pmatrix} 1/4 & -1/4 \\ -1/4 & 1/4 \end{pmatrix}\begin{pmatrix} 3/2 \\ 0 \end{pmatrix} = \begin{pmatrix} 3/14 \\ -3/14 \end{pmatrix}.$$

This is the actual fine-to-coarse discrepancy, not a fitted approximation. The matrix kills $\bar{\lambda}$, recording the common-scale null direction. Its ℓ^1 error is $3/7$.

For a family of likelihoods, states may be grouped by equality of their *whole likelihood profile*. The family $(3, 3, 1, 1)$ and $(1, 2, 1, 2)$ separates all four states even though either likelihood alone permits a coarser grouping. The common refinement is therefore the discrete partition. The theorem's one-packet exception still remains.

> **Read this as:** The defect factors through exactly the conditional evidence information missing from the declared coarse model. A new observation family can require a finer partition.

BAYESIAN FLOW / COMPOSITION WITHOUT LINEARITY

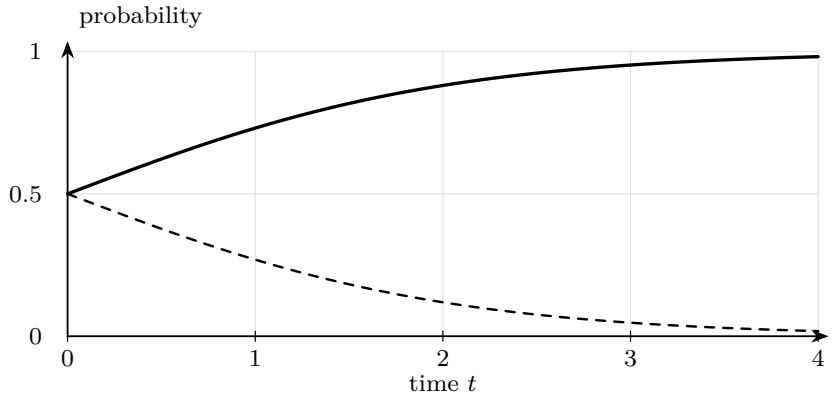
A nonlinear semigroup of odds

Main text: Theorems 7.1 and 9.1. Likelihood composition and the replicator field.

Let the score vector be $g = (1, 0)$ and start at $(1/2, 1/2)$. The likelihood at time t is $(e^t, 1)$, so the first posterior coordinate is

$$p_0(t) = \frac{e^t}{1 + e^t}, \qquad p_1(t) = \frac{1}{1 + e^t}.$$

The odds p_0/p_1 grow exponentially, while the probabilities remain inside the simplex. At $t = \log 2$ the posterior is exactly $(2/3, 1/3)$.



The solid and dashed curves sum to one. The curvature is caused by the shared evidence denominator, not by a nonlinear likelihood multiplication law.

Two properties that must not be conflated

If $a > 0$ multiplies the odds, the first-coordinate map is

$$\phi_a(x) = \frac{ax}{1 + (a - 1)x}.$$

Direct algebra gives $\phi_a(\phi_b(x)) = \phi_{ab}(x)$. Thus $a = e^t$ gives a nonlinear semigroup. Nevertheless, $\phi_3(1/2) = 3/4$, while the mean of $\phi_3(0)$ and $\phi_3(1)$ is $1/2$. Superposition fails.

Differentiation of the normalized flow yields the centered score equation

$$\dot{p}_i = (g_i - \langle g, p \rangle) p_i.$$

For the displayed model, $\dot{p}_0 = p_0(1 - p_0)$. The negative average-score term is what keeps the total derivative zero. Omitting the derivative of the normalizer would not preserve mass.

Likelihood composition also preserves the evidence identity: the first evidence times the second conditional evidence equals the combined evidence. Normalization cancels in the posterior, but that factor still matters when comparing data models.

> **Read this as:** A semigroup is a composition law in time. It does not imply that the maps are linear or that they preserve mixtures.

HIDDEN MARKOV FILTERING / TIME ORDER

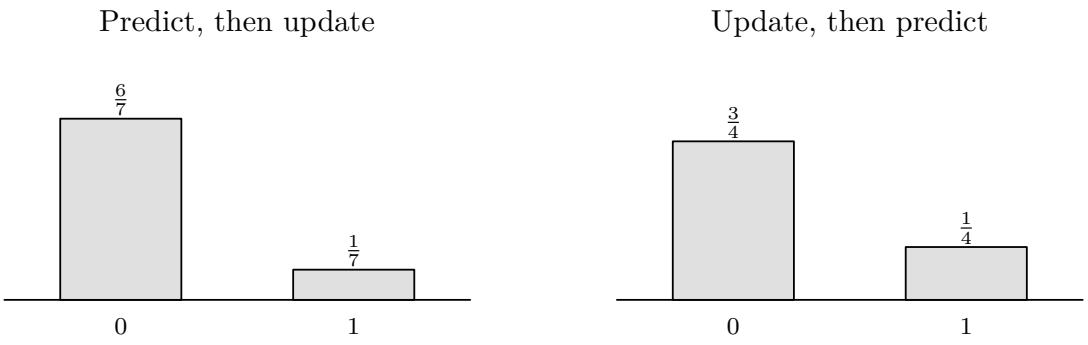
Prediction and observation do not commute

Main text: Proposition 9.3. Linear prediction followed by a positive-evidence update.

Use the two-state kernel K_2 from V – 2, the initial distribution e_0, and likelihood $(1, 1/2)$. Predicting first gives $(3/4, 1/4)$. Its likelihood-weighted vector is $(3/4, 1/8)$, with evidence $7/8$, so the posterior is $(6/7, 1/7)$.

Updating first leaves e_0 unchanged. Predicting afterward gives $(3/4, 1/4)$ instead. The order of operations expresses which time the observation concerns.

Same prior e_0, transition K_2, and likelihood $(1, 1/2)$



Both routes produce valid probability vectors. Their difference is therefore not a normalization bug. It is a different conditioning problem.

Keep an unnormalized path alongside the filter

For a sequence of observations, let $D_r = \operatorname{diag}(\lambda^{(r)})$. The forward recursion is

$$\widehat{p}_r = K p_{r-1}, \qquad p_r = D_r \widehat{p}_r / Z_r.$$

Its unnormalized companion is

$$v_r = D_r K \cdots D_1 K p_0, \qquad p_r = \frac{v_r}{\mathbf{1}^{\mathsf{T}} v_r}, \qquad \mathbf{1}^{\mathsf{T}} v_r = \prod_{j=1}^{r} Z_j.$$

The product of evidences records the normalization lost at each individual step. The four-state computation on V – 26 checks both paths independently, including their coarse versions.

Here $D_r K$ means an observation after prediction. In the Feynman–Kac example on V – 20, departure-state weighting has forward operator $K D_w$. Both conventions are legitimate once their meanings are fixed, but they cannot be substituted for one another.

> **Read this as:** Write the chronological sequence before multiplying matrices. A product can be normalized correctly and still answer the wrong time-ordered question.

BETA MODEL / POSTERIOR AND EVIDENCE

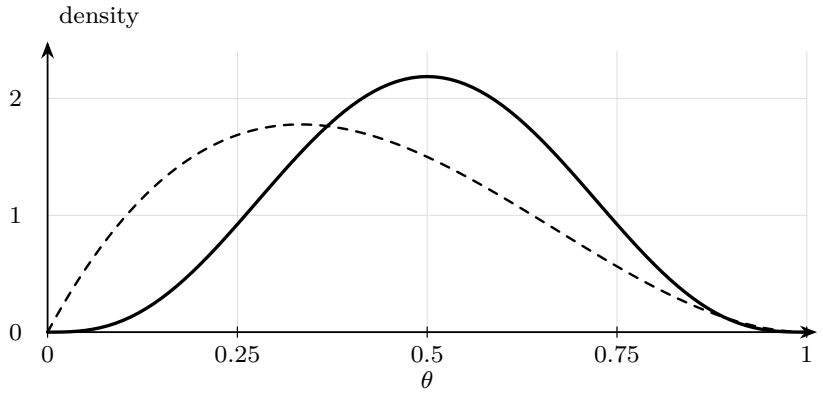
Conjugacy tracks exponents

Main text: Example 9.2. A finite-parameter conjugate family, not a universal posterior model.

For a small exact example, take the Beta$(2, 3)$ density

$$\rho(\theta) = 12\theta(1-\theta)^2, \qquad 0 \le \theta \le 1.$$

Observe a specified ordered sequence with two successes and one failure. Its likelihood is $\theta^2(1-\theta)$. Multiplication adds exponents, so the posterior belongs to Beta$(4, 4)$ with density $140\theta^3(1-\theta)^3$.



Prior: Beta$(2, 3)$

Posterior: Beta$(4, 4)$

ordered data: $3/35$
count data: $9/35$

Same posterior,
different evidence.

The dashed prior is asymmetric. The solid posterior is symmetric with mean $1/2$. Polynomial integration verifies the normalizers exactly; the curves are sampled drawings of those polynomials.

The observation convention changes the evidence

The integral of the unnormalized posterior is

$$\int_0^1 12\theta^3(1-\theta)^3\,d\theta = \frac{3}{35}.$$

Dividing by $3/35$ gives the displayed normalized density. If the observation is instead the *count* of two successes in three trials, there are $\binom{3}{2} = 3$ compatible ordered sequences. The count likelihood is therefore $3\theta^2(1-\theta)$ and its evidence is $9/35$.

The extra factor does not depend on θ, so it cancels from the posterior. The two experiments thus have the same posterior distribution but different data probabilities. Evidence is not determined by the final normalized density alone.

The exact check also integrates both densities to one and integrates θ against the posterior to obtain $1/2$. This demonstrates one conjugate family in elementary terms. It does not show that arbitrary likelihoods preserve that family or that all posterior information can be compressed into two exponents.

> **Read this as:** Conjugacy gives a closed parameter update when a specified likelihood respects a specified family. Preserve the evidence as well as the normalized posterior.

FRACTIONAL DIFFUSION / NEW COUPLINGS

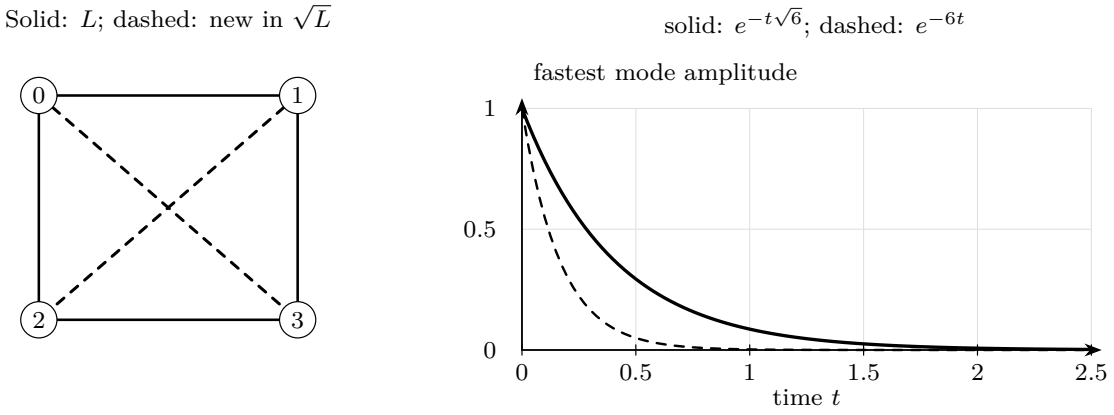
The modes stay; their clocks change

Main text: Theorem 10.1. A positive self-adjoint finite model and its spectral powers.

Return to the four-state generator on V − 4 and put $L = -Q$. It is real symmetric and nonnegative, with orthogonal projectors P_0, P_1, P_2, P_3 and eigenvalues $0, 2, 4, 6$. Its square root is

$$L^{1/2} = \sqrt{2}P_1 + 2P_2 + \sqrt{6}P_3, \qquad (L^{1/2})^2 = L.$$

Both heat evolutions retain these modes. Ordinary diffusion multiplies mode j by e^{-2jt}; square-root diffusion multiplies it by $e^{-\sqrt{2j}\,t}$. The constant mode is unchanged in both.



The highest mode decays at rates 6 and $\sqrt{6}$. The graph beside the curves shows the additional couplings in the fractional operator. Curves display the explicit formulas, not an empirical fit.

Fractional does not mean entrywise

Although $L_{03} = 0$, its spectral square root satisfies

$$(L^{1/2})_{03} = \frac{\sqrt{6} - \sqrt{2} - 2}{4} < 0.$$

Thus $-L^{1/2}$ permits a direct transition between states 0 and 3, which were not neighbors for Q. Taking the square root of each matrix entry would neither produce this value nor square back to L.

The exact block aggregation remains compatible in this example:

$$CL^{1/2} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} C.$$

The coarse matrix is the positive square root of $-Q_c$. This is a concrete naturality identity, checked on the full matrices. It is not a claim that an arbitrary partition commutes with every fractional construction.

The computation uses a nonnegative self-adjoint operator and a specified spectral branch. General nonnormal matrices, unbounded domains, and positivity of other fractional evolutions need their own hypotheses.

> **Read this as:** Fractional powers alter modal decay rates. Reassembling those modes can introduce long-range spatial interactions even when the original generator was sparse.

FRACTIONAL DEFECTS / NEAR ZERO VERSUS AWAY FROM ZERO

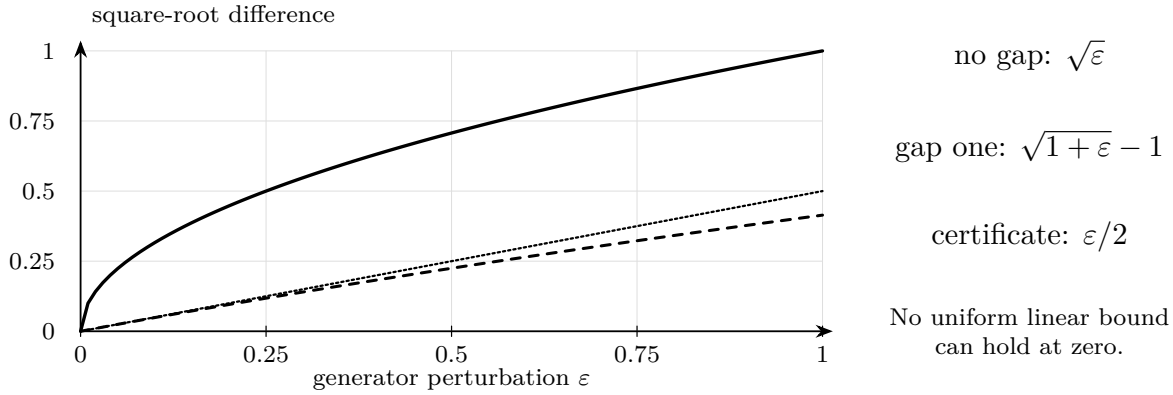
A gap changes the error scale

Main text: Theorem 10.3 and Corollary 10.4. Separate generator error from fractional error.

The function $a \mapsto \sqrt{a}$ changes most rapidly near zero. A scalar model already shows why a spectral gap appears in fractional-defect estimates:

$$|\sqrt{\varepsilon} - \sqrt{0}| = \sqrt{\varepsilon}, \qquad \sqrt{1+\varepsilon} - 1 = \frac{\varepsilon}{\sqrt{1+\varepsilon}+1} \leq \frac{\varepsilon}{2}.$$

The same input difference has a square-root scale at zero and a linear bound above a fixed positive lower bound.



The upper curve compares 0 with ε. The lower curve compares 1 with $1+\varepsilon$. The dotted line $\varepsilon/2$ bounds the latter. All plotted values come from the displayed scalar formulas.

Read an operator bound with its hypotheses

For positive self-adjoint finite matrices $A, B \geq mI$ and $0 < \alpha < 1$, the manuscript's gapped estimate has the form

$$\|CA^{\alpha} - B^{\alpha}C\| \leq \alpha m^{\alpha-1} \|CA - BC\|.$$

It concerns an intertwining defect, not just two eigenvalue lists. With $C = 1$, $A = 4$, $B = 9$, $m = 4$, and $\alpha = 1/2$, the generator defect has size 5, the fractional defect has size 1, and the bound is $5/4$.

The common zero mode of a Markov generator cannot simply be included in a positive-gap estimate. One must either work on an appropriate gapped subspace, handle the zero mode separately, or use the gapless estimate. In particular, a good estimate on each finite level need not have a constant uniform in the refinement level.

On V – 27, the sequences $1/n^2$ and $1/n$ turn this change in error scale into a concrete distinction between summable and nonsummable defects.

> **Read this as:** A fixed gap is quantitative information. It prevents the derivative of the fractional function from becoming singular along the comparison.

SUBORDINATION / A DISTRIBUTION OF ORDINARY TIMES

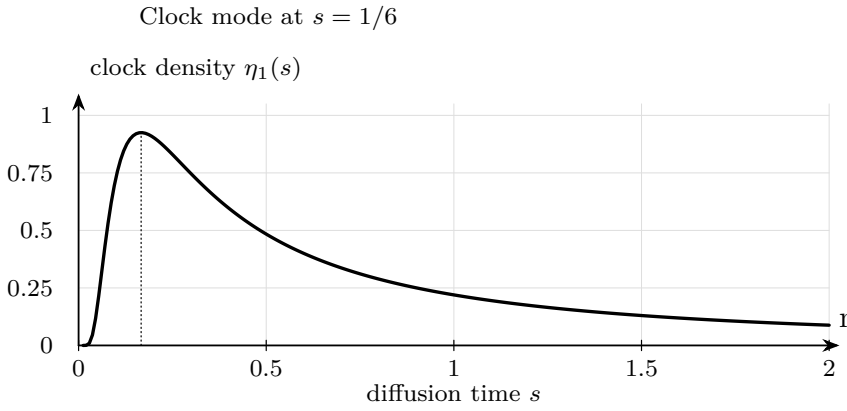
A random diffusion clock

Main text: Theorem 10.2. The Poisson subordination formula for the finite positive model.

Square-root evolution can be written as an average of ordinary diffusion over a positive clock variable s:

$$e^{-t\sqrt{L}} = \int_0^\infty \eta_t(s) e^{-sL}\, ds, \qquad \eta_t(s) = \frac{t}{2\sqrt{\pi}} s^{-3/2} e^{-t^2/(4s)}, \quad t > 0.$$

Here t is the external time; s is the ordinary diffusion time being mixed. They are not interchangeable variables.



$$\int_0^\infty \eta_t(s)\, ds = 1$$

$$\int_0^\infty s\eta_t(s)\, ds = \infty$$

A probability clock need not have a finite mean.

The density η_1 peaks at $s = 1/6$ but has a long right tail. The drawing ends at $s = 2$; it does not claim that the missing tail has zero mass.

Check the normalizer before interpreting a density

Under $s = t^2/(4u^2)$, reversing the integration limits gives

$$\eta_t(s)\,|ds| = \frac{2}{\sqrt{\pi}} e^{-u^2}\, du.$$

Its integral over $u > 0$ is one, so η_t is a probability density. On each eigenmode of L, the operator formula reduces to the scalar Laplace identity

$$\int_0^\infty \eta_t(s) e^{-\lambda s}\, ds = e^{-t\sqrt{\lambda}}.$$

The manuscript supplies that scalar identity and its spectral assembly. The appendix separately checks the change of variables and normalization; sampled curves are not used to certify an improper integral.

A normalized density need not have a finite mean. For large s, the exponential factor tends to one and $\eta_t(s)$ is proportional to $s^{-3/2}$. Consequently $s\eta_t(s)$ has a nonintegrable $s^{-1/2}$ tail. The clock has infinite mean even though its total mass is one.

> **Read this as:** Subordination is a weighted mixture of diffusion times. Replacing the clock distribution by a single average time is not justified, and here that average is infinite.

DIRICHLET GRID / THE MISSING FACTOR OF h

A Green kernel is an inverse with units

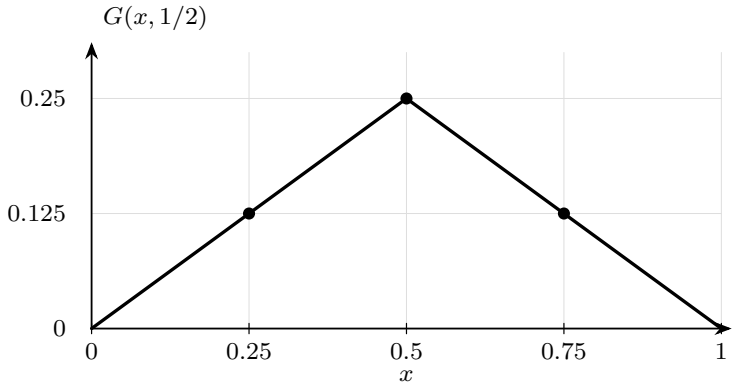
Main text: Theorem 11.1. Exact nodal Green reconstruction for this one-dimensional stencil.

Take three interior nodes $x_i = i/4$ on $[0, 1]$, with zero boundary values and $h = 1/4$. The discrete negative second derivative is

$$L_h = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

For the continuous Dirichlet problem, the Green function is $G(x, y) = \min(x, y) - xy$. At these nodes the exact comparison is

$$L_h^{-1} = h\,[G(x_i, x_j)]_{i,j=1}^{3} = \frac{1}{64} \begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}.$$



$64L_h^{-1}$, $h = 1/4$

3	2	1
2	4	2
1	2	3

$L_h^{-1} = hG_{\text{nodes}}$

The tent is the central column of the continuous Green kernel. It has nodal values $(1, 2, 1)/8$; its discrete inverse column is $(1, 2, 1)/32$, smaller by the factor h.

A forcing vector is not a quadrature-weighted measure

For sampled forcing f_j, the finite solution is

$$u_i = \sum_j (L_h^{-1})_{ij} f_j = h \sum_j G(x_i, x_j) f_j.$$

The factor h converts samples into the quadrature weights appropriate to this formula. Dropping it multiplies the computed solution by four on the displayed grid.

A discrete unit coordinate vector and a discretized unit point mass use different normalizations. For $f = e_2$, the result is the second inverse column. For $f = e_2/h$, it is the unscaled tent sampled at the nodes. Either convention can be useful, but it must be specified before comparing answers.

Exact checks multiply L_h by the proposed inverse at $1, 3, 7$, and 15 interior nodes. The general identity is the manuscript theorem. These finite checks do not extend it to arbitrary meshes, dimensions, or boundary conditions.

> **Read this as:** An inverse kernel carries a measure convention. A correct-looking tent can give a wrong solution if the grid weight is omitted.

FINITE DIFFERENCES / A QUARTIC TEST PROBLEM

Convergence with a nonzero exact error

Main text: Theorems 11.1–11.2. Exact nodal inversion and the scale of low-mode errors.

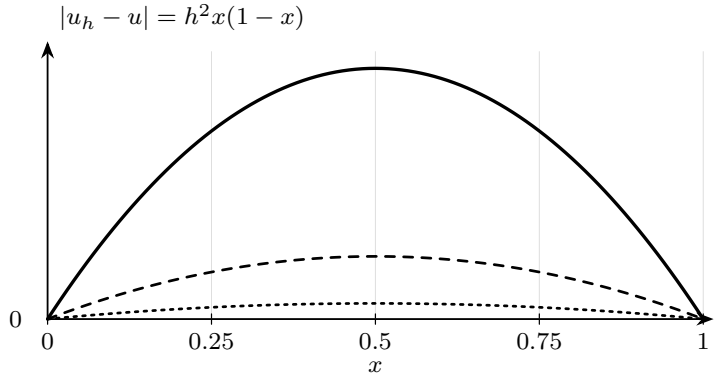
Use $u(x) = x - x^4$, which vanishes at both endpoints and solves $-u'' = 12x^2$. Its centered second difference is not exact:

$$L_h u|_{\text{nodes}} = 12x_i^2 + 2h^2.$$

Since the stencil differentiates the quadratic $x(1-x)$ exactly, the solution u_h to the sampled forcing $12x_i^2$ satisfies

$$\boxed{u_h(x_i) - u(x_i) = -h^2 x_i(1-x_i).}$$

This gives both the sign and size of the error, rather than merely a log–log slope.



grid spacing h	maximum error
$\frac{1}{4}$	$\frac{1}{64}$
$\frac{1}{8}$	$\frac{1}{256}$
$\frac{1}{16}$	$\frac{1}{1024}$

The three curves show magnitudes of the exact nodal errors for $h = 1/4, 1/8, 1/16$. Lines between grid points are visual guides, not an additional continuous numerical solution. The error is negative on every interior node.

One refinement divides this error by four

All three grids include $x = 1/2$, where the maximum absolute nodal error is $h^2/4$. It equals 1/64, 1/256, and 1/1024, respectively. Exact rational matrix inversion verifies these values independently of the closed error formula.

The manuscript also compares the discrete eigenvalues

$$\lambda_{k,h} = 4h^{-2}\sin^2(k\pi h/2)$$

with $(k\pi)^2$, with error at most $(k\pi)^4 h^2/12$. For a fixed mode k, the bound tends to zero quadratically. It does not claim a uniform small error for modes whose index grows like $1/h$.

The nodal Green identity is exact, while the discretization of the differential equation still has a truncation error. These statements concern different comparisons and do not contradict one another.

> **Read this as:** An exact inverse of the discrete operator does not make that operator equal to the differential operator. Keep algebraic inversion and continuum approximation separate.

FEYNMAN–KAC / DEPARTURE-STATE WEIGHTS

Eight paths, one weighted expectation

Main text: Theorem 12.1. A finite path sum with its time convention displayed.

Let $K_2 = \frac{1}{4}\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, start at state 0, use weights $w = (1, 1/2)$, and terminal observable $f = (2, 1)$. For three transitions, there are eight possible state sequences after the fixed initial state. We evaluate

$$\mathbb{E}_0\left[\prod_{r=0}^{2} w(X_r)\, f(X_3)\right].$$

Each path contributes its transition probability times its three departure weights times its terminal value.

Path	probability	weight	terminal	contribution
0000	27/64	1	2	27/32
0001	9/64	1	1	9/64
0010	3/64	1/2	2	3/64
0011	9/64	1/2	1	9/128
0100	3/64	1/2	2	3/64
0101	1/64	1/2	1	1/128
0110	3/64	1/4	2	3/128
0111	9/64	1/4	1	9/256

Total probability = 1; weighted terminal expectation = 311/256.

Every possible path is included. The probability column sums to one. The contribution column sums to 311/256, agreeing exactly with the transfer calculation.

Check which side of the transition gets weighted

The backward operator on observables is $D_w K_2^{\mathsf{T}}$. Therefore the expectation is the zeroth coordinate of

$$(D_w K_2^{\mathsf{T}})^3 f.$$

Its forward dual is $K_2 D_w$: weight the mass at the departing state, then move it. Weighting arrival states instead gives $D_w K_2$, a different expression. The exact tests deliberately check that the two conventions do not agree in this example.

The answer 311/256 exceeds one, but it is not a probability. The terminal observable can equal two. With $0 \le w_i \le 1$, the weighted expectation is bounded by the maximum terminal value, not necessarily by one.

This calculation certifies a finite three-step identity for a stated model. The continuous-time theorem uses the corresponding infinitesimal generator and potential, with the direction of action kept consistent.

> **Read this as:** A Feynman–Kac weight changes the value assigned to each path. Its location relative to a transition is part of the model, not a typographical choice.

SCHUR–FESHBACH / RETAINED RESOLVENT

Elimination makes a frequency-dependent term

Main text: Theorem 13.1. A two-state Schur complement with the inverse checked directly.

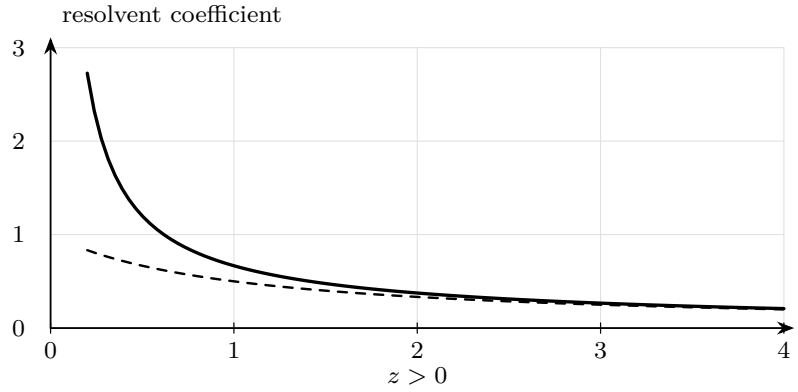
The two-state evolution from V−3 has generator

$$A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Retain the first coordinate and eliminate the second. For z away from the relevant poles, the retained entry of $(zI - A)^{-1}$ is

$$R_x(z) = \left[z + 1 - \frac{1}{z+1} \right]^{-1} = \frac{z+1}{z(z+2)}.$$

The term $\Sigma(z) = 1/(z+1)$ is the hidden coordinate's self-energy.



exact: $\dfrac{z+1}{z(z+2)}$

omit hidden state: $\dfrac{1}{z+1}$

The pole at zero survives through hidden feedback.

The full inverse gives a rational response with two poles, at 0 and −2. Its partial fractions are $R_x(z) = \frac{1}{2}z^{-1} + \frac{1}{2}(z+2)^{-1}$. The hidden intermediate inverse has its own excluded point $z = -1$.

An eliminated variable has not disappeared

If the coupling to the second coordinate were simply deleted, the retained resolvent would be $(z+1)^{-1}$. That has a different pole and a different long-time limit. The correction is frequency dependent, so it cannot be represented by one constant scalar generator on the retained coordinate.

The Schur expression first requires $z + 1 \neq 0$, as well as invertibility of the full matrix. Its simplified rational form has a removable intermediate singularity at $z = -1$, where the retained entry of the full inverse is well-defined and equals zero. Simplification does not retroactively make the eliminated block invertible there.

For $z > 0$, the Laplace transform of $(1 + e^{-2t})/2$ is exactly $R_x(z)$. The next page derives the time-domain equation producing that response.

> **Read this as:** The Schur complement is exact elimination, not independent dynamics on a smaller state space. Hidden evolution returns as a z-dependent self-energy.

EXACT RETAINED DYNAMICS / NOT A SCALAR SEMIGROUP

Hidden state returns as memory

Main text: Theorem 13.2. The retained-state memory identity, including hidden initial data.

Write the same system as

$$x' = -x + y, \qquad y' = x - y.$$

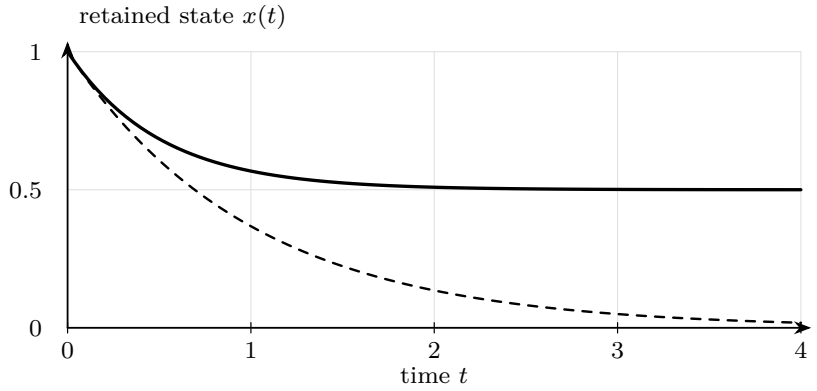
Solving the second equation first gives

$$y(t) = e^{-t}y(0) + \int_0^t e^{-(t-r)}x(r)\,dr.$$

Substitution is exact:

$$x'(t) = -x(t) + e^{-t}y(0) + \int_0^t e^{-(t-r)}x(r)\,dr.$$

The hidden initial state produces a separate forcing term. It must be retained even when the memory kernel is known.



exact: $\dfrac{1 + e^{-2t}}{2}$

naive deletion: e^{-t}

$K(t) = e^{-t}$
returns hidden excursions.

For $(x(0), y(0)) = (1, 0)$, the exact retained coordinate is $(1 + e^{-2t})/2$. The dashed curve e^{-t} solves the equation obtained by deleting the hidden coordinate rather than eliminating it.

Projection can destroy the semigroup law

Let $R(t) = (1 + e^{-2t})/2$, the response to this fixed initial preparation. At $t = \log 2$,

$$R(t) = \frac{5}{8}, \qquad R(2t) = \frac{17}{32}, \qquad R(2t) - R(t)^2 = \frac{9}{64}.$$

The full two-state propagator is a semigroup, but this retained scalar response is not. Restarting it as a scalar model silently resets the hidden coordinate, which generally carries information from the first interval.

The appendix verifies the full differential equation, initial condition, integral memory equation, and Laplace resolvent in this model. Its kernel is e^{-t}, whose transform is the self-energy on V – 21.

> **Read this as:** Exact reduction retains a record of the eliminated state through convolution and initial forcing. A memoryless approximation must justify the information it removes.

NONDIAGONAL HIDDEN DYNAMICS / POLYNOMIAL EXPONENTIALS

A repeated pole remembers a Jordan chain

Main text: Theorem 13.2 and Section 15. Functional calculus must retain nilpotent data.

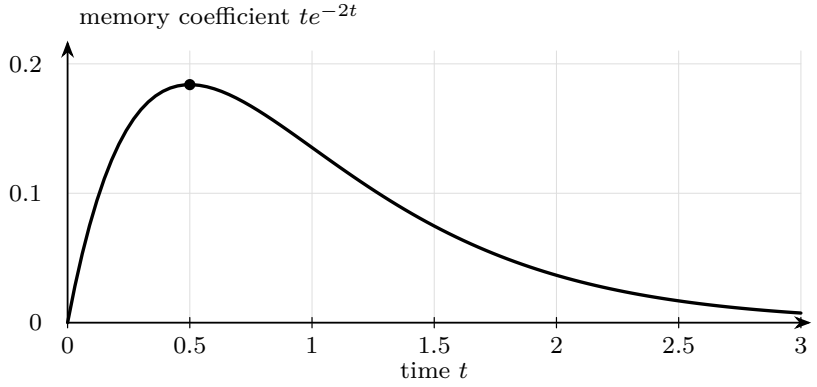
Now use a two-dimensional hidden block

$$J = -2I + N, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

Choose the entrance column $V = (0,1)^{\mathsf{T}}$ and exit row $U = (1,0)$. The hidden exponential is not $e^{-2t}I$:

$$e^{tJ} = e^{-2t}(I + tN), \qquad K_{\mathrm{mem}}(t) = Ue^{tJ}V = te^{-2t}.$$

The nilpotent link carries an input from the second hidden coordinate to the first.



$$J = -2I + N, \quad N^2 = 0$$

$$e^{tJ} = e^{-2t}(I + tN)$$

$$\widehat{K}(z) = \frac{1}{(z+2)^2}$$

A repeated pole carries a jet.

The memory kernel starts at zero, becomes positive, and peaks at $t = 1/2$. A rule that retained the eigenvalue -2 but discarded the Jordan chain would predict the zero kernel for these entrance and exit maps.

The transfer function sees pole order

The exact transform is

$$U(zI - J)^{-1}V = \frac{1}{(z+2)^2} = \int_0^\infty te^{-(z+2)t}\,dt, \qquad \operatorname{Re} z > -2.$$

The repeated pole and the polynomial factor t are the same information in two domains. For a longer nilpotent chain, higher powers of t occur with factorial denominators.

A zero value $K_{\mathrm{mem}}(0)$ does not make the entire kernel zero. Nor do stable eigenvalues alone specify every transient contribution: entrance maps, exit maps, and nilpotent structure matter. Here the full kernel is nevertheless elementary, so its peak and transform can be checked without numerical fitting.

This page is an explicit finite realization. It does not show that an arbitrary continuum memory kernel admits a finite-dimensional representation of this form.

> **Read this as:** Repeated eigenvalues may carry directional information through Jordan chains. Keep the finite jet of the functional calculus, not just its values on the spectrum.

PHASE-TYPE SURVIVAL / TWO SUCCESSIVE STAGES

A zero initial hazard can still lead to certain exit

Main text: Theorem 14.1. Survival, density, hazard, and moments have distinct meanings.

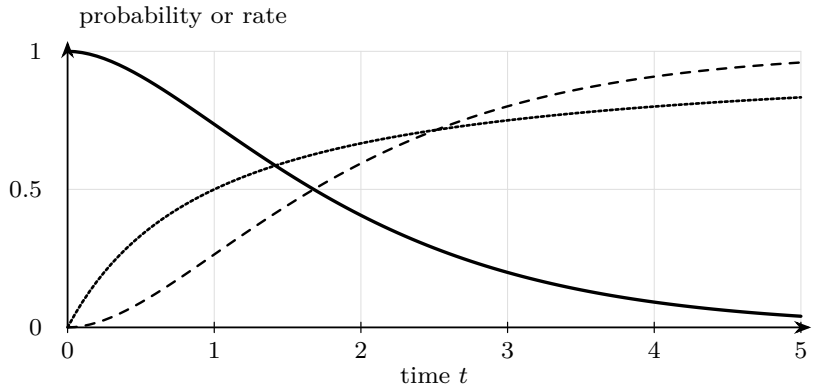
Let a transient process move through two successive rate-one stages, then leave the system. With column probabilities, the transient generator and starting vector are

$$T = \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix}, \qquad p(0) = e_0.$$

Thus $p(t) = e^{-t}(1, t)^{\mathsf{T}}$. Its total remaining mass, exit density, and hazard are

$$S(t) = (1 + t)e^{-t}, \qquad f(t) = te^{-t}, \qquad h(t) = \frac{t}{1 + t}.$$

The initial hazard is zero because an exit needs two stages, not one.



solid: survival $S(t)$

dashed: absorbed mass

dotted: hazard $h(t)$

$h(0) = 0$, but eventual exit has probability one.

Survival falls from one to zero, while absorbed mass $1 - S$ rises to one. The dotted hazard is an instantaneous rate, not a probability, even though it shares the displayed numerical scale. Time is measured in inverse stage-rate units.

Missing transient mass is absorbed mass

The column sums of T are 0 and -1. The exit-rate row is therefore $(0, 1)$, and $f(t) = (0, 1)p(t)$. Adding an absorbing state restores a three-state probability vector with total mass one.

Exact integration gives

$$\int_0^\infty f(t)\, dt = 1, \qquad \mathbb{E}[\tau] = 2, \qquad \mathbb{E}[\tau^2] = 6, \qquad \mathrm{Var}(\tau) = 2.$$

Equivalently, the mean is $\mathbf{1}^{\mathsf{T}}(-T)^{-1}e_0$. A single exponential waiting time with mean two would have constant hazard $1/2$ and variance four, so matching the mean alone would not reproduce this distribution.

The hazard formula is used where $S(t) > 0$. The model has that property for every finite $t \geq 0$. No division by an already vanished survival mass is required.

> **Read this as:** Survival tracks mass, density tracks exits per unit time, and hazard conditions on survival. Equal means or equal initial rates do not identify a waiting-time law.

HOLOMORPHIC CALCULUS / A RECTANGULAR COMPARISON

Naturality also transports derivatives

Main text: Theorem 15.1. Exact intertwining passes to the specified functional calculus.

Consider a fine operator, a coarse operator, and a rectangular map

$$A_f = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 9 \end{pmatrix}, \qquad A_c = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \qquad C_r = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Direct multiplication gives $C_r A_f = A_c C_r$. For $F(z) = 1/(1+z)$, which is defined at both spectra, compute the matrices by inversion:

$$F(A_f) = \begin{pmatrix} 1/3 & -1/9 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/10 \end{pmatrix}.$$

A_f

2	1	0
0	2	0
0	0	9

$F(A_f)$ for $F(z) = (1+z)^{-1}$

$\frac{1}{3}$	$-\frac{1}{9}$	0
0	$\frac{1}{3}$	0
0	0	$\frac{1}{10}$

$$F(2) = 1/3$$

$$F'(2) = -1/9$$

The dashed block is $F(A_c)$, not just $F(2)I$.

The discarded eigenvalue 9 contributes $1/10$ only on the fine side. On the retained Jordan block, the off-diagonal entry is $F'(2) = -1/9$, not zero. The rectangular comparison preserves that entry.

Why spectral values are not enough

On the retained block $A_c = 2I + N$, with $N^2 = 0$, one has

$$F(A_c) = F(2)I + F'(2)N = \tfrac{1}{3}I - \tfrac{1}{9}N.$$

Hence $C_r F(A_f) = F(A_c) C_r$. Replacing the block by $F(2)I$ would lose the nonzero derivative term and no longer give the actual inverse of $I + A_c$.

For polynomials, naturality follows by transporting every operator power. The manuscript's finite holomorphic statement uses finite Taylor data at each primary block and the corresponding nilpotent orders. This example verifies a rational instance with an actual inverse rather than asserting that arbitrary scalar substitutions are meaningful.

The same mechanism explains why the Jordan memory kernel on V – 23 contains te^{-2t}. Exponentiation also has nonzero derivatives at repeated spectral values.

> **Read this as:** A functional calculus acts on operators, including their nilpotent parts. Exact intertwining transports that full action, not merely a list of scalar eigenvalue images.

END-TO-END MODEL / TWO OBSERVATIONS

One complete coarse filter

Main text: Proposition 9.3 with Theorems 5.1 and 8.2. Markov closure and likelihood sufficiency are separate.

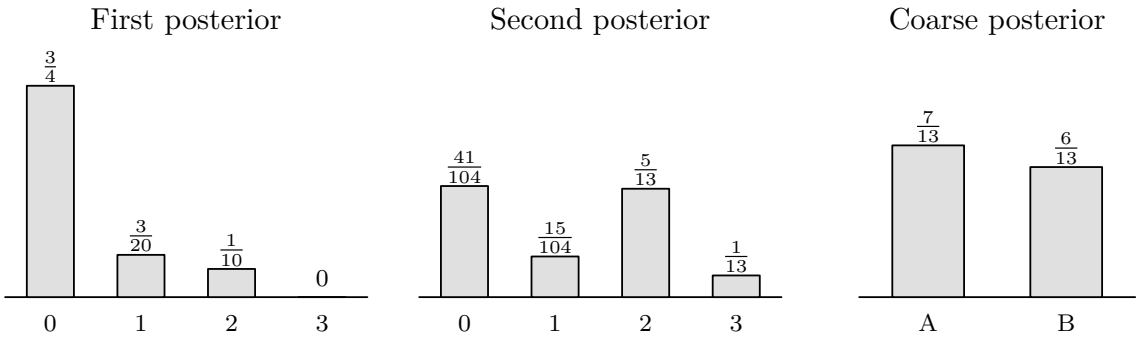
Use the four-state kernel K and aggregation C from V – 5, starting at $p_0 = e_0$. Observe two likelihood vectors constant inside both packets:

$$\lambda^{(1)} = (3/4, 3/4, 1/4, 1/4), \qquad \lambda^{(2)} = (1/2, 1/2, 1, 1).$$

Predict, then update at each observation. The fine calculation gives

$$p_1 = \tfrac{1}{20}(15, 3, 2, 0)^{\mathsf{T}}, \qquad p_2 = \tfrac{1}{104}(41, 15, 40, 8)^{\mathsf{T}}.$$

Evidence $Z_1 = 5/8$, $Z_2 = 13/20$, and $Z_1 Z_2 = 13/32$



Within-packet probabilities change during the two observations, but the packet totals agree exactly with an independent two-state filter. The second coarse posterior is $(7/13, 6/13)$.

Check the normalizers, not only the final vector

The first and second predictive evidences are

$$Z_1 = \frac{5}{8}, \qquad Z_2 = \frac{13}{20}, \qquad Z_1 Z_2 = \frac{13}{32}.$$

The unnormalized product $D_2 K D_1 K p_0$ has mass $13/32$ and normalizes to p_2. Because the likelihood entries here lie in $[0, 1]$, they can serve as probabilities of the specified observations. This makes the product their joint evidence in the stated model.

Two distinct equalities justify the coarse computation. First, $CK = K_c C$ transports prediction. Second, the likelihood is constant on each packet, so $CD_r = \overline{D}_r C$ transports observation weighting. Their composition preserves both the posterior packet totals and the evidence factors.

The coarse filter does not recover the within-packet shape. If a later observation distinguishes states inside a packet, those missing probabilities can become necessary, as the counterexample on V – 10 shows.

> **Read this as:** Exact filtering at a coarse level needs both dynamical lumpability and observation sufficiency. Agreement under one step does not excuse testing the other.

REFINEMENT LIMITS / WHAT THE FINITE MODELS DO NOT PROVE

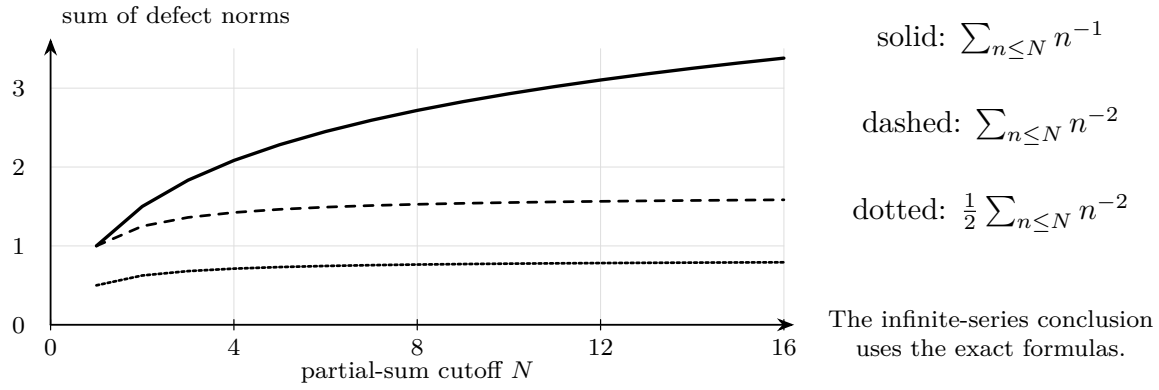
Small at each level is not summable across levels

Main text: Corollary 10.5 and Section 17. Summability and uniform control are additional data.

For each $n \geq 1$, compare the two scalar operators $A_n = 0$ and $B_n = 1/n^2$ through the identity. Their ordinary defects have sizes $1/n^2$, while their square-root defects have sizes $1/n$:

$$\sum_n |A_n - B_n| < \infty, \qquad \sum_n |A_n^{1/2} - B_n^{1/2}| = \infty.$$

These are independent scalar comparisons, not a constructed inverse system. They isolate an error-scale obstruction without claiming a coherent refinement tower.



Partial sums through $n = 16$ illustrate the contrast. The convergence and divergence statements follow from the displayed series, not from the finite plot. A common positive gap gives the third, summable upper bound.

A uniform hypothesis can change the conclusion

Replace the pair by $A_n = 1$ and $B_n = 1 + 1/n^2$. Then

$$\left|B_n^{1/2} - A_n^{1/2}\right| = \frac{1/n^2}{\sqrt{1 + 1/n^2} + 1} \leq \frac{1}{2n^2}.$$

This is the elementary form of the uniform-gap mechanism on V – 16. Without such information, transferring a summable generator-defect budget through a fractional function can fail.

The formal release keeps three original research questions separate: realization of broader transfer semigroups, uniform stochastic-refinement control, and latent packet closure. The exact finite examples here do not settle their existence, domain, topology, or uniformity requirements.

Likewise, exact hidden-state elimination only yields the finite memory models that were supplied. It does not establish a finite latent realization for every memory law. A complete list of matrix identities can still leave an infinite-dimensional approximation problem untouched.

> **Read this as:** A finite example can expose a missing hypothesis. It cannot supply a projective-limit construction, a uniform constant, or a continuum realization merely by passing at several levels.

READER'S GUIDE / CALCULATION, DRAWING, AND FORMAL CONTRACT

How to use the evidence

Source release: Paper VI v0.5.0, Checked Fifth Block. The original manuscript and proof boundaries are preserved.

This appendix is an explanatory layer. The existing 34-page manuscript remains unchanged in the combined edition, including its figures and formal-status appendix. Source lines beneath these titles link back to the relevant manuscript pages in that edition.

The chosen release records all 33 scoped results complete, with 341 supporting theorem declarations in 47 proof modules and a saved audit of 782 namespace declarations. Those are inherited verification records, not a new Lean run and not counts of historically novel results. The three original conjectures retain their stated status.

Read each kind of evidence at its actual strength

Evidence	What is checked here	What it does not establish
Exact matrices	Stochasticity, intertwining, projector identities, inverse formulas, and specific defects.	The same properties for arbitrary partitions or unbounded operators.
Rational algebra	Posterior/evidence identities, polynomial integrals, nodal errors, and complete finite path sums.	An unstated observation model or a continuum convergence theorem.
Symbolic formulas	The displayed exponentials, Laplace transforms, derivatives, and finite spectral powers under stated conditions.	A general-purpose formal proof of every analytic identity.
Vector drawings	Reproducible geometry and sampled curves from the explicit models.	Numerical quadrature certificates or an exhaustive infinite-time experiment.
Saved Lean records	The original contract, source files, dependency pins, and earlier verification logs.	Fresh kernel verification of this editorial addition.

Reproduce a calculation before adapting it

The visual-appendix tree contains the mathematical models, drawing generator, and exact-example verifier. Its README records their paths and execution commands. The verifier currently checks 204 exact or symbolic assertions and rejects 11 deliberately false controls. Its JSON record names every check and its finite scope.

The finite families are explicit: three-step Feynman–Kac paths are fully enumerated, selected rational priors are checked exhaustively within their stated denominator bound, and Green inverses are checked on four named grids. No finite count is used as a substitute for a general theorem.

To change a model, specify the state convention, partition, likelihood meaning, topology, and domain first. Then rerun the calculations and inspect the regenerated drawings. A preserved formula with changed assumptions is not automatically a preserved result.

> **Read this as:** Use the appendix to see why a hypothesis matters and to compute a small case. Use the manuscript and its formal map for the scope of the general claim.